



Epigenetic encoding, heritability and plasticity of glioma transcriptional cell states

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Single-cell RNA sequencing has revealed extensive transcriptional cell state diversity in cancer, often observed independently of genetic heterogeneity, raising the central question of how malignant cell states are encoded epigenetically. To address this, here we performed multiomics single-cell profiling—integrating DNA methylation, transcriptome and genotype within the same cells—of diffuse gliomas, tumors characterized by defined transcriptional cell state diversity. Direct comparison of the epigenetic profiles of distinct cell states revealed key switches for state transitions recapitulating neurodevelopmental trajectories and highlighted dysregulated epigenetic mechanisms underlying gliomagenesis. We further developed a quantitative framework to directly measure cell state heritability and transition dynamics based on high-resolution lineage trees in human samples. We demonstrated heritability of malignant cell states, with key differences in hierarchical and plastic cell state architectures in IDH-mutant glioma versus IDH-wild-type glioblastoma, respectively. This work provides a framework anchoring transcriptional cancer cell states in their epigenetic encoding, inheritance and transition dynamics.

Single-cell RNA sequencing (scRNA-seq) of human tumors provides a powerful means to systematically interrogate the diversity of malignant and normal cell states. Recent studies have highlighted transcriptional cell state diversity across tumor types that is often independent of genetic clonal heterogeneity^{1–3}. Thus, tumors are composed of admixtures of cells that differ in central phenotypes^{1,4–7}, prompting several key questions. For example, how are transcriptional cell states encoded epigenetically? How heritable are malignant cell states? Further, what are the transition dynamics between cell states? While exploration of these central aspects of cancer cell states has begun in model organisms using artificial constructs for lineage tracing^{8–12}, these questions remain largely unexplored in primary patient samples.

Human gliomas serve as an instructive model to address these questions, as cell state diversity is an important disease hallmark of both IDH-mutant (IDH-MUT) glioma and IDH-wild-type glioblastoma (GBM), with malignant cells recapitulating trajectories of neural development^{13–16}. Stemness-to-differentiation diversity is central to the glioma stem-cell (GSC) model, which posits that stem-like cells are uniquely capable of self-renewal, tumor propagation and preferential resistance to therapy^{17–19}. Recent scRNA-seq profiling of gliomas provided high-resolution mapping of cell state diversity and offered additional granularity to the GSC model by revealing multiple transcriptionally defined cell states related to

neurodevelopmental cell types, which are in part independent of intratumoral genetic diversity^{7,16,20–26}. Yet, while cellular states can be precisely delineated by scRNA-seq, transcriptional information provides only a snapshot of the current state of the cell; therefore, glioma cell state heritability and transition dynamics are not readily discernable. Indeed, while malignant cell states may be propagated epigenetically^{27–30}, the epigenetic underpinning of glioma cellular states is still largely unknown.

This question is of clinical relevance as heritable expression programs may be related to non-genetic mechanisms of therapy resistance in cancer^{5,6}. Increased plasticity allowing for both differentiation and dedifferentiation may also offer a mechanism by which tumors could replenish their stem-cell compartment under therapeutic pressure. Attempts at addressing the dynamics of cell state transitions in glioma samples with stand-alone scRNA-seq modalities (for example, by RNA velocity³¹) have generated conflicting results³², suggesting that additional technological and analytical breakthroughs are required. To address these questions, we applied joint capture of transcriptional, genetic and epigenetic information at single-cell resolution³³ to primary diffuse gliomas. We leveraged this approach to increase the resolution of single-cell identification of copy number alterations (CNAs), demonstrate significant DNA methylation intratumoral heterogeneity (ITH), and reveal the epigenetic encoding, heritability and plasticity of cell states in glioma.

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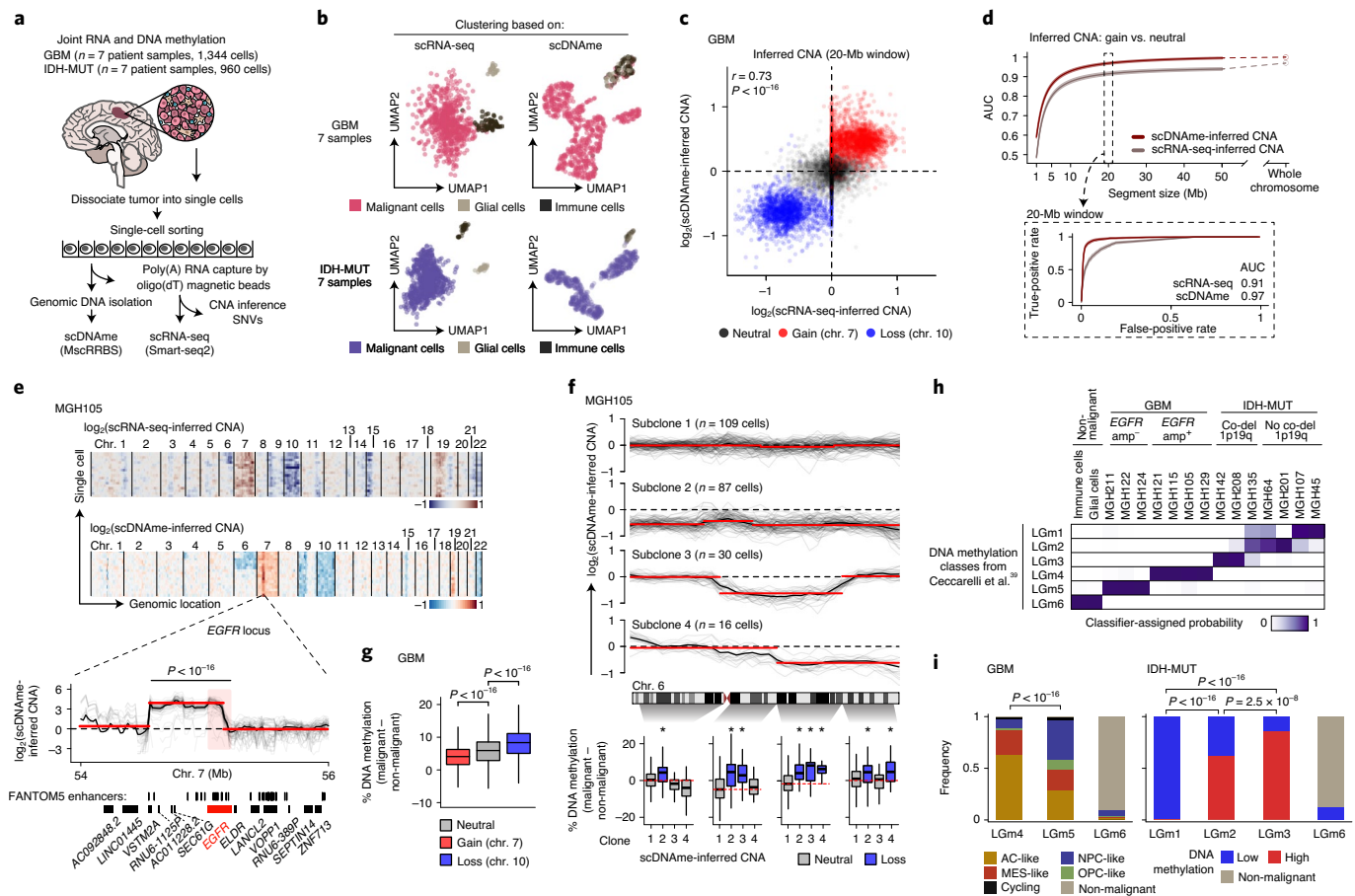


Fig. 1 | Multiomics single-cell sequencing of primary human gliomas reveals intratumoral DNA methylation heterogeneity. **a**, Joint methylomics and transcriptomics analysis applied to seven GBM and seven IDH-MUT glioma samples. **b**, UMAP plots of single cells that passed quality control based on data from scRNA-seq (left; GBM, $n = 937$; IDH-MUT, $n = 809$) or scDNAme (right; GBM, $n = 867$; IDH-MUT, $n = 718$). **c**, CNA inference by scDNAme (y axis) versus scRNA-seq (x axis) in 20-Mb windows. Pearson's correlation coefficient is indicated. **d**, Performance of CNA inference by scDNAme (red line) and scRNA-seq (gray line) in correctly classifying regions of chromosome gain versus neutral regions, as assessed by the AUC of the receiver operating characteristic (ROC) curve at different genomic window resolutions. Inset, ROC curves at 20-Mb resolution. 95% confidence intervals were generated using bootstrapping. **e**, Top, representative example (MGH105) of CNA inference by scRNA-seq and scDNAme. Rows correspond to cells, clustered by overall CNA pattern. Bottom, CNA inference by scDNAme centered at the *EGFR* locus. CNA profiles for individual cells are shown in gray, with the mean per sample shown in black. Red lines represent CNA segments identified by circular binary segmentation analysis. **f**, Top, CNA inference by scDNAme at chromosome 6 showing distinct genetic subclones within the same tumor (MGH105). Color legend as in **e**. Bottom, CpG methylation changes at four regions of chromosome 6 when comparing the DNA methylation level of individual cells in each subclone to baseline. * $P < 0.05$. **g**, Percentage of CpG methylation change at regions with copy number gain (chromosome 7) or loss (chromosome 10) and neutral regions (chromosome 1) when comparing DNA methylation levels for individual GBM cells to baseline across all GBM samples. **h**, Heat map of probability assignment for pseudo-bulk DNA methylation profiles (based on MscRRBS) of malignant cells across all GBM and IDH-MUT glioma samples and non-malignant cells (defined in **b**) to previously described LGM classes³⁹ based on a multinomial logistic regression classifier (Methods). Amp, amplification; co-del, co-deletion. **i**, Proportion of all single cells (defined in **b**) assigned to previously described DNA methylation LGM classes³⁹. P values were determined by two-sided Mann-Whitney U test (**e-g**) or Fisher's exact test (**i**). Boxplots represent the median and bottom and upper quartiles; whiskers correspond to 1.5 times the interquartile range.

Results

High-resolution CNA mapping by single-cell multiomics. We profiled viable cells enriched for CD45⁺ cells from GBM ($n = 7$) and IDH-MUT glioma ($n = 7$) primary patient samples with multi-modality single-cell sequencing of DNA methylation (scDNAme; by multiplexed single-cell reduced-representation bisulfite sequencing (MscRRBS)), scRNA-seq (Smart-seq2; ref. ³⁴) and targeted genotyping³³ (Fig. 1a, Extended Data Figs. 1 and 2, and Supplementary Tables 1 and 2). After quality control, we obtained a mean of 113 cells per sample (range, 28–339 cells), with DNA methylomes with a mean $\pm$ s.e.m. of $198,345 \pm 4,307$ unique CpGs per cell and transcriptomes with a mean $\pm$ s.e.m. of $6,348 \pm 43$ genes per cell (Supplementary Table 3),

comparable to results with stand-alone full-length scRNA-seq^{7,21,22}. We then separated malignant cells from non-malignant cells on the basis of clustering of either gene expression or DNA methylation data (Fig. 1b and Extended Data Fig. 2c). Non-malignant cells expressed either typical oligodendrocytic markers (for example, *PLP1*) or myeloid cell markers (for example, *CD14*) (Extended Data Fig. 3a).

To orthogonally validate malignant versus non-malignant classification, we identified CNAs within each cell on the basis of coverage depth imbalance in the DNA methylation data (Extended Data Figs. 1a and 2a). CNA inference by scDNAme enabled robust detection of amplifications and deletions in malignant cells, including the hallmark chromosome 7 gain and chromosome 10 loss in GBM and

chromosome 1p/19q co-deletion in IDH-MUT oligodendroglioma (IDH-O) (Extended Data Figs. 1a and 2a). While CNA inference by scDNAme correlated with CNA inference by scRNA-seq^{7,21} (Pearson's $r=0.73$; Fig. 1c), direct comparison^{7,21} at clonal CNAs (identified by bulk whole-exome sequencing with matched samples) (Extended Data Fig. 3b) showed that scDNAme-based CNA inference afforded greater resolution (Fig. 1d, Extended Data Fig. 3c and Supplementary Table 4) and enabled detection of focal amplifications of oncogenes (for example, *EGFR*, encoding epidermal growth factor receptor) and their neighboring enhancers³⁵ (Fig. 1e and Extended Data Fig. 3d–f).

Higher-resolution scDNAme-based CNA inference further revealed the presence of genetic subclones in both GBM and IDH-MUT tumors (Extended Data Figs. 1a and 2a). For example, we identified distinct genetic subclones marked by either complete or partial chromosome 6 loss in four spatially distinct regions sampled from the same GBM tumor (MGH105) (Fig. 1f, top). Notably, copy number loss was associated with increased methylation, such that DNA methylation levels increased specifically in the chromosome 6 segments lost in each subclone ([6p25–6p11], [6q12–6q15], [6q16–6q23.2] and [6q23.3–6q27]; Fig. 1f, bottom). This pattern was observed more broadly, with increased DNA methylation with copy number loss (for example, loss of chromosome 10 in GBM or chromosomes 1p/19q in IDH-O tumors) and decreased DNA methylation with copy number gain (for example, gain of chromosome 7 in GBM or chromosomes 7/8 in IDH-MUT tumors) across patient samples (Fig. 1g and Extended Data Fig. 3g–i). While such an association between CNAs and subtle DNA methylation changes (<5% on average) has previously been observed in bulk samples³⁶, the underlying mechanism remains unclear and may be related to recruitment of DNA methyltransferases (DNMT1 and DNMT3B) and Polycomb family members (SIRT1 and EZH2) at the chromosomal breaks that lead to CNAs³⁷. The observed anticorrelation between copy number and DNA methylation may serve as a mechanism that amplifies gene expression changes due to CNAs³⁸.

Single-cell DNA methylation analysis reveals significant DNA methylation ITH. Diffuse gliomas have been classified into six distinct tumor subtypes (LGm1–LGm6) by bulk DNA methylation analysis³⁹. LGm1–LGm3 are enriched for IDH-MUT tumors and show genome-wide hypermethylation, while LGm4–LGm6 are enriched for GBM tumors. We hypothesized that the scDNAme data might reveal ITH in these bulk DNA methylation profiles, that

is, that each IDH-MUT tumor might be composed of an admixture of the LGm1–LGm3 DNA methylation subtypes, while each GBM tumor might span the LGm4–LGm6 subtypes. To test this hypothesis, we trained a classifier, robust across DNA methylation platforms, on 932 glioma samples from The Cancer Genome Atlas (TCGA)^{40,41} profiled with the 450K methylation array and recovered the expected bulk DNA methylation subtypes, achieving a mean accuracy of 0.94 in fivefold cross-validation. When this classifier was applied to pseudo-bulk DNA methylation profiles (based on MscRRBS) of malignant cells in our samples, it assigned each sample to its expected DNA methylation subtype, with IDH-MUT pseudo-bulk DNA methylation profiles classified as LGm1, LGm2 or LGm3 depending on their 1p/19q co-deletion status and GBM samples resolved into either LGm4 or LGm5 depending on their *EGFR* amplification status (Fig. 1h and Extended Data Fig. 3d–f). Notably, pseudo-bulk analysis of non-malignant glial and immune cells classified them into LGm6 (Fig. 1h), a subtype found in 77 of the 932 TCGA gliomas and associated with either GBM or pilocytic astrocytoma-like gliomas, suggesting that the tumor microenvironment may contribute to bulk subtype assignments to LGm6.

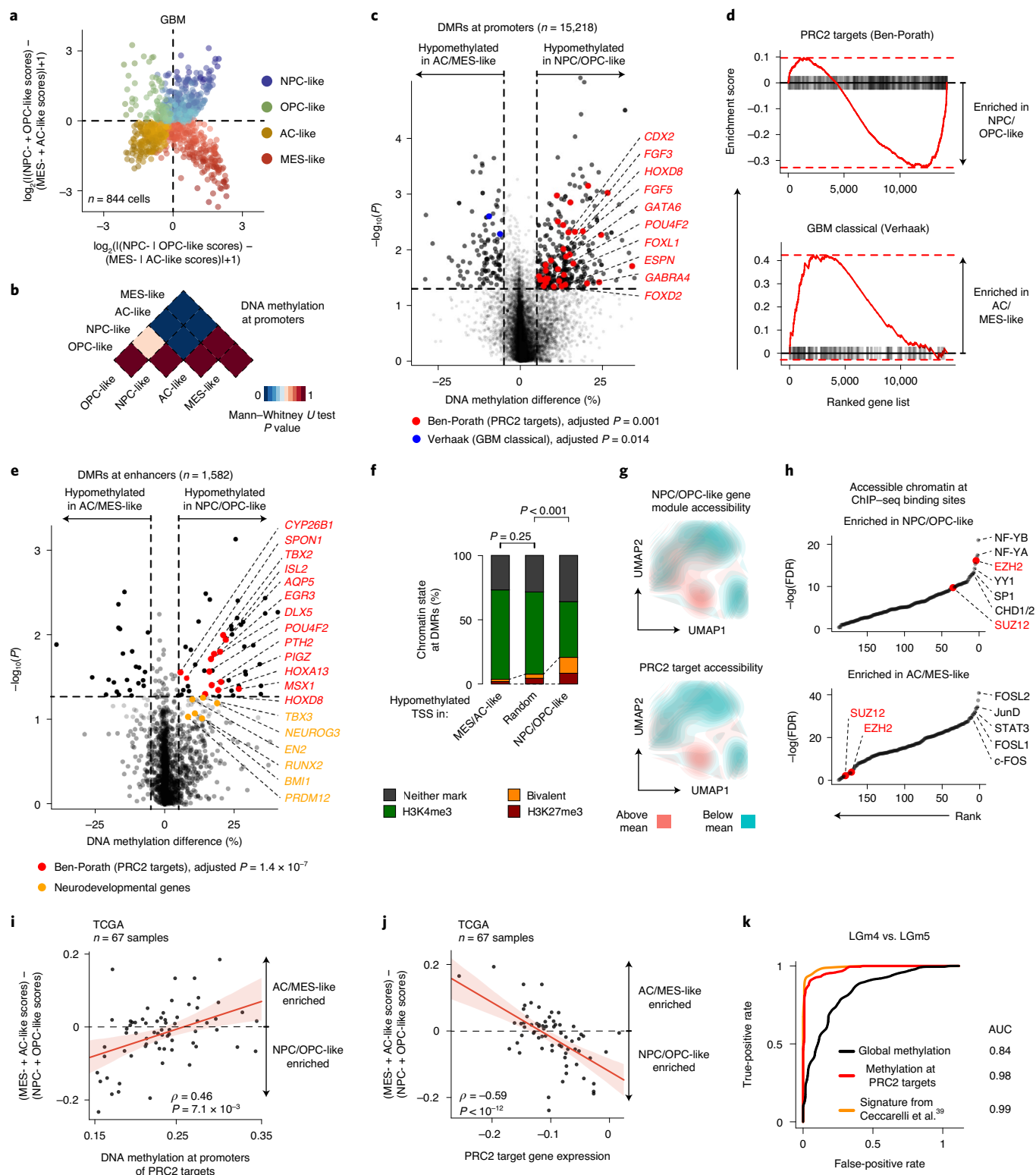
We then scored each glioma single cell to the six tumor subtypes (LGm1–LGm6; Methods) and observed that single cells within individual IDH-MUT tumors spanned the LGm1–LGm3 subtypes, while single cells within individual GBM tumors spanned the LGm4 and LGm5 subtypes (Extended Data Figs. 1c and 2d,e). Such ITH in DNA methylation subtypes is important to recognize, as bulk DNA methylation profiling is increasingly being used for clinical classification of brain tumors⁴². In IDH-MUT tumors, no correlation with cellular states⁷ was detected, but instead we found an association with genome-wide DNA methylation levels (Fisher's exact test, $P < 2.5 \times 10^{-8}$; Fig. 1i and Extended Data Fig. 2d–f), as previously observed in bulk DNA methylation profiles³⁹. By contrast, in GBM, ITH in the LGm4 and LGm5 DNA methylation subtypes correlated with recently defined GBM cellular states based on the expression of defining gene modules in matching scRNA-seq profiles²¹ (correlation of LGm4 with AC- and MES-like cell states and LGm5 with NPC- and OPC-like cell states; cell state definitions below; Fisher's exact test, $P < 10^{-16}$) (Fig. 1i and Extended Data Fig. 1c,d).

GBM stem-like cells exhibit PRC2 target hypomethylation. To define the distinct DNA methylation profiles of glioma cell states, we first classified glioma cells on the basis of expression of gene modules and cell cycle programs previously defined in scRNA-seq

Fig. 2 | PRC2 target DNA methylation is a key switch in the differentiation of malignant GBM cells. **a**, Two-dimensional representation of cellular states across seven GBM samples ($n=844$ malignant-only cells that passed scRNA-seq quality control). **b**, Heat map of P values obtained when comparing mean CpG methylation at promoters (TSS ± 1 kb) between GBM cellular states ($n=706$, cells in **a** with matched scDNAme data). **c**, Volcano plot of differentially methylated promoters between the NPC/OPC-like and AC/MES-like GBM cellular states. Promoters ($n=459$) with an absolute mean DNA methylation difference of greater than 5% and a P value below 0.05 were defined as differentially methylated. Genes correlated with the classical TCGA GBM subtype⁴⁵ (blue) and genes corresponding to PRC2 targets⁴⁶ (red) are highlighted (BH FDR-adjusted permutation-based $P < 0.05$). **d**, Enrichment score plots (reflecting whether a gene set is over-represented at the top or bottom of the ranked list of genes used in **c**; $n=15$, 218 genes) for gene sets enriched at hypomethylated promoters in NPC/OPC-like (top) or AC/MES-like (bottom) cells. **e**, Volcano plot of differentially methylated enhancers in comparison of NPC/OPC-like and AC/MES-like GBM cellular states. Putative gene targets⁵¹ of hypomethylated enhancers in stem-like cells that are PRC2 targets⁴⁶ are labeled in red. Key neurodevelopmental genes are highlighted in orange. **f**, Proportion of chromatin states at hypomethylated promoters in GBM AC/MES-like cells (defined in **c**), randomly sampled promoters (1,000 promoters sampled) and hypomethylated promoters in GBM stem-like cells (defined in **c**). **g**, UMAP plots of scATAC-seq GBM data⁵⁵ (sample SF11956) overlaid with density plots showing chromatin accessibility of genes belonging to NPC/OPC-like gene modules (top) and PRC2 targets⁴⁶ (bottom). **h**, Comparison of rank (by P value) in the enrichment of open chromatin at transcription factor-binding sites ($n=188$) between NPC/OPC-like (top) and AC/MES-like (bottom) cells. PRC2 subunits are highlighted in red. **i**, Spearman's rank-order correlation between mean DNA methylation at the promoters of PRC2 targets⁴⁶ and RNA differentiation score (defined as the difference in gene module scores between AC/MES-like and NPC/OPC-like cellular states) for 67 TCGA GBM samples^{40,41}. A linear regression line (red) with its 95% confidence interval is shown. **j**, Same as in **i**, for PRC2 target⁴⁶ gene expression and RNA differentiation score. **k**, Comparison of performance, as assessed by ROC curve, in correctly classifying bulk TCGA GBM samples^{40,41} to DNA methylation glioma subtypes (LGm4 or LGm5)³⁹ using 1,300 previously defined CpG sites³⁹, mean global DNA methylation and mean PRC2 target⁴⁶ DNA methylation. P values were determined by two-sided Mann–Whitney U test (**b**), generalized linear model (**c,e**), permutation test (**f**), BH FDR-adjusted hypergeometric test (**h**) or Spearman's rank-order correlation (**i,j**).

data^{7,21} (Methods). GBM samples exhibited four malignant cell states, spanning stem/progenitor-like cells (neural progenitor-like (NPC-like) cells and oligodendrocyte progenitor-like (OPC-like) cells) and more differentiated states associated with astrocyte-like (AC-like) or mesenchymal-like (MES-like) programs (Fig. 2a and Supplementary Table 5), with varying representation across samples and cell cycle expression (Extended Data Fig. 1b), as previously observed²¹.

Comparison of promoter DNA methylation between transcriptional cell states revealed that, while stem-like cells were markedly different from differentiated-like cells, smaller differences were present in promoter DNA methylation levels within stem-like cells or within differentiated-like cells (Fig. 2b, Extended Data Fig. 4a and Supplementary Table 6). These data suggest that these pairs of cell states are more closely related to each other and that regulatory mechanisms other than DNA methylation, such as interaction



with the tumor microenvironment, may drive certain state transitions⁴³. In line with cross-talk between GBM cells and immune cells driving MES-like cell state transitions^{43,44}, immune response-related genes were found to be upregulated in MES-like cells (Benjamini–Hochberg (BH) false-discovery rate (FDR)-adjusted $P < 0.05$; Extended Data Fig. 4b and Supplementary Table 7).

We thus focused our analysis on comparison of DNA methylation profiles between stem-like and more differentiated-like states, identifying 459 promoter differentially methylated regions (DMRs) (Fig. 2c, Extended Data Fig. 4c,d and Supplementary Table 6). Hypomethylated promoters in AC- and MES-like cells were enriched for genes correlated with the ‘classical’ TCGA GBM subtype (TCGA-CL)⁴⁵, in line with the enrichment of AC-like cells in TCGA-CL (BH FDR-adjusted permutation-based $P < 0.05$; Fig. 2c,d, Extended Data Fig. 4e and Supplementary Table 6).

By contrast, we identified Polycomb repressive complex 2 (PRC2) targets⁴⁶ as hypomethylated in NPC- and OPC-like cells as compared to AC- and MES-like cells (BH FDR-adjusted permutation-based $P < 0.05$; Fig. 2d, Extended Data Fig. 4e,f and Supplementary Table 6). These hypomethylated PRC2 targets were enriched for HOX (for example, *HOXD8*, *HOX11* and *HOXA6*) and homeobox (for example, *CDX2* and *POU4F2*) genes, as well as for transcription factors (for example, *GATA5*, *GATA6*, *FOXJ1* and *LHX2*) and growth factors (for example, *FGF3*–*FGF5*) (BH FDR-adjusted Fisher’s exact test, $P < 0.05$; Supplementary Table 6), previously reported to have a role in the epigenetic regulation of stemness in GBM⁴⁷. Notably, NPC- and OPC-like cells exhibited DNA hypomethylation of PRC2 targets as compared to AC- and MES-like cells even within GBM samples from the same patient (Extended Data Fig. 4g–i), suggesting that PRC2 target DNA hypomethylation is a key determinant of stem-like GBM cell states^{48,49}. This was further confirmed when using chromatin immunoprecipitation and sequencing (ChIP-seq) maps⁵⁰ for the PRC2 subunits EZH2 and SUZ12 (Mann–Whitney U test, $P < 0.0001$; Extended Data Fig. 4j). We similarly defined enhancer DMRs and found that the putative gene targets⁵¹ of hypomethylated enhancers in stem-like cells were also enriched for PRC2 targets⁴⁶ (Fig. 2e and Supplementary Table 6). As direct cross-talk between PRC2 and DNA methylation has been reported^{52,53}, these data suggest that DNA methylation marks cell states through its interaction with PRC2 and its ability to catalyze the addition of H3K27me3 marks.

To explore the link between DNA methylation and histone marks, we interrogated the differentially methylated promoters for enrichment of histone marks associated with non-overlapping regulatory functions⁴⁷. While hypomethylated promoters in AC- and MES-like cells were predominantly marked by histone modifications associated with active transcription (H3K4me3, H3K27ac and H3K36me3), hypomethylated promoters in NPC- and OPC-like cells were enriched in bivalent (H3K4me3 + H3K27me3) chromatin (permutation-based $P < 0.001$; Fig. 2f and Extended Data Fig. 5a–e), suggesting that PRC2 complex activity may result in poised transcription at these gene promoters⁵⁴. Indeed, the PRC2 subunit EZH2 and its targets⁴⁶ were found to be upregulated (>2-fold increase) in NPC- and OPC-like cells in comparison to AC- and MES-like cells (Extended Data Fig. 5f,g and Supplementary Table 7).

To further validate the association between the stem-like states and PRC2 activity, we reanalyzed data from GBM single-cell assay for transposase-accessible chromatin sequencing (scATAC-seq)⁵⁵. GBM cells formed clusters associated with the four core malignant cellular states described by scRNA-seq (Extended Data Fig. 5h). Gene expression activity inferred from scATAC-seq open chromatin (Methods) revealed a positive correlation between PRC2 target accessibility and the NPC- and OPC-like cellular states in single cells (hypergeometric test, $P = 0.0015$; Fig. 2g and Extended Data Fig. 5i). Similarly, intersecting open chromatin with ChIP-seq maps revealed that binding sites for the PRC2 subunits EZH2 and SUZ12

were among the most enriched in NPC- and OPC-like cells as compared to AC- and MES-like cells (Fig. 2h).

To examine this association in a larger sample cohort, we leveraged 67 GBM samples from TCGA with matched bulk RNA-seq and 450K methylation profiles^{40,41}. In line with our model, we found a positive correlation between the DNA methylation of PRC2 targets⁴⁶ and glioma differentiation (Fig. 2i and Extended Data Fig. 5j), as well as an anticorrelation between the expression of PRC2 targets⁴⁶ and glioma differentiation (Fig. 2j). These data confirm that PRC2 targets not only are hypomethylated but also show greater expression in stem-like cells. We note that these findings are consistent with the suppressive role of PRC2, as its targets showed lower gene expression than non-PRC2 targets across all GBM samples. However, the degree of repression was stronger in tumors enriched for differentiated-like cell states, where these gene promoters also underwent silencing through DNA methylation (Mann–Whitney U test, $P < 0.0001$; Extended Data Fig. 5k). As expected, PRC2 target promoter DNA methylation was lower in LGm5 cells (enriched for NPC- and OPC-like cells) than in LGm4 cells (enriched for AC- and MES-like cells) (Extended Data Fig. 6a). TCGA bulk glioma DNA methylation profiles recapitulated this finding with lower PRC2 target DNA methylation in LGm5 tumors than in LGm4 tumors (Extended Data Fig. 6b,c). In fact, using just mean PRC2 target DNA methylation as a single feature in the classifier separated bulk glioma DNA methylation subtypes (LGm4 and LGm5)³⁹ with comparable accuracy as the multinomial logistic regression classifier (area under the curve (AUC) of 0.98 versus 0.99, respectively; Fig. 2k), suggesting that PRC2 target DNA methylation underlies the classification of GBM tumors by bulk DNA methylation.

Collectively, these data show that DNA methylation of PRC2 targets is a critical feature of GBM cell differentiation. This epigenetic encoding of glioma supports the parallels between glioma differentiation and physiological neurodevelopment where stemness is also marked by PRC2 target hypomethylation⁵⁶. Maintaining PRC2 targets in a hypomethylated state in glioma stem-like states may thus preserve their stemness potential and allow their reactivation in response to stimuli.

Aberrant epigenetic and transcriptional mechanisms in IDH-MUT gliomas. In line with previous reports, IDH-MUT malignant cells were found to be differentiated along the astrocytic (AC-like) or oligodendrocytic (OC-like) glial lineages, with a subpopulation of undifferentiated cells associated with an NPC-like expression program²³ (Extended Data Fig. 2b and Supplementary Table 5). Cells with cell cycle expression signatures were enriched in this latter subpopulation, supporting a model in which stem-like cells are primarily responsible for fueling the growth of IDH-MUT tumors⁷ (Extended Data Fig. 2b). In contrast to GBM, differentially methylated promoters in comparisons of stem-like cells with AC- and OC-like cells in IDH-MUT samples were not enriched for PRC2 targets (Extended Data Fig. 7a–g and Supplementary Table 8). In addition, we did not observe significant enrichment of bivalent and repressive chromatin marks at hypomethylated promoters in stem-like cells as compared to AC- and OC-like cells (Extended Data Fig. 7h–l), suggesting that different epigenetic patterning is at play in the maintenance of stemness in IDH-MUT gliomas.

Mutated IDH produces 2-hydroxyglutarate (2HG), an oncometabolite and a competitive inhibitor of the TET family of 5-methylcytosine hydroxylases⁵⁷. TET enzymes oxidize 5-methylcytosines to promote demethylation, and deficiency in TET activity may lead to increased DNA methylation, primarily at regulatory elements^{58–61}. Indeed, DNA methylation levels were highest in IDH-MUT cells as compared to GBM and non-malignant cells at gene promoters (Mann–Whitney U test, $P < 10^{-16}$; Fig. 3a). Comparison of GBM and IDH-MUT samples revealed that

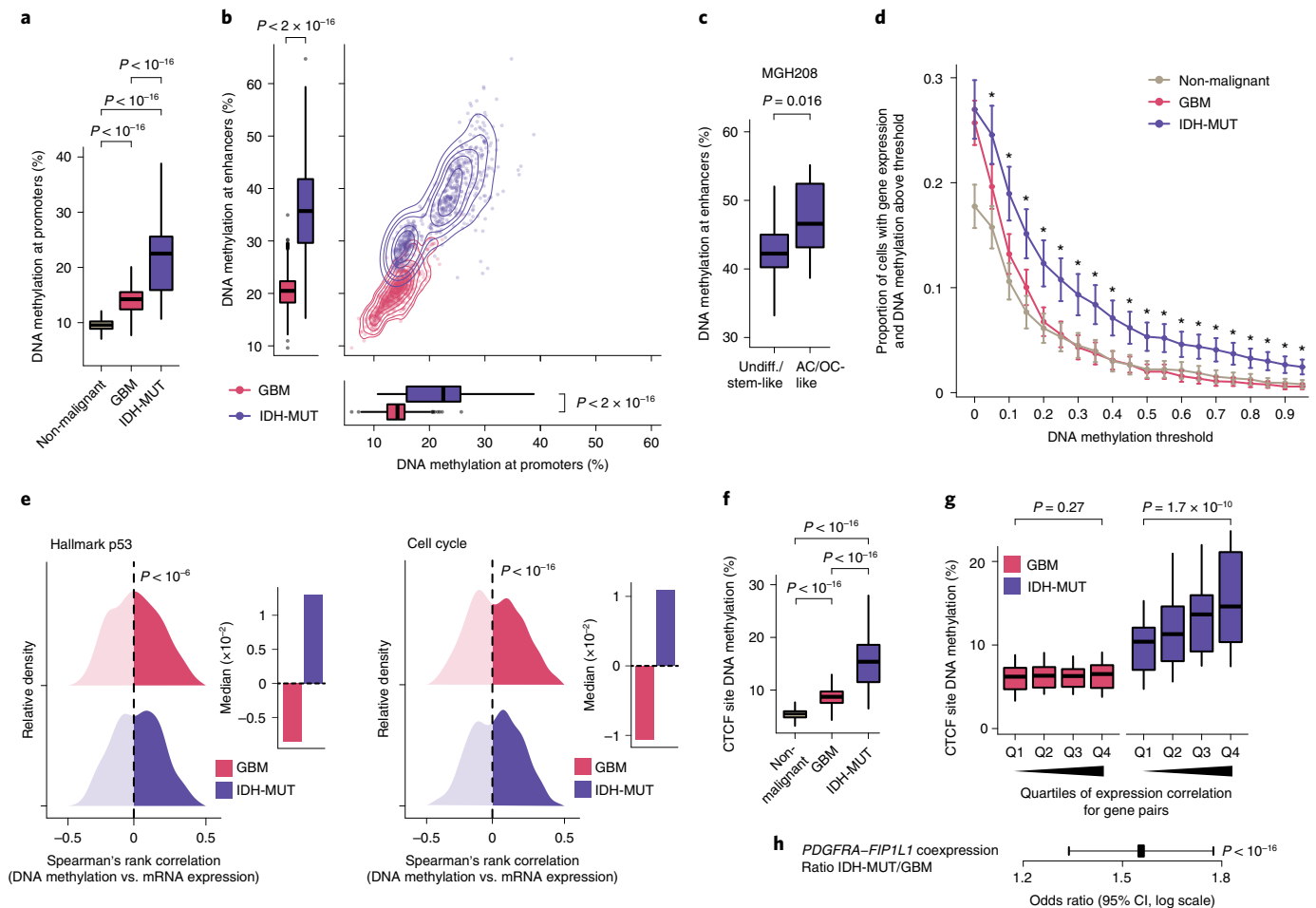


Fig. 3 | Increased enhancer DNA methylation, decoupling of promoter methylation-expression relationship and disruption of CTCF insulation define the IDH-MUT epigenome. **a**, Mean CpG methylation at promoters (TSS ± 1 kb) comparing non-malignant cells ($n=148$) with GBM ($n=765$) and IDH-MUT ($n=670$) malignant cells. **b**, Mean CpG methylation at promoters versus FANTOM5 enhancers for GBM ($n=765$) and IDH-MUT ($n=670$) malignant cells. **c**, Mean CpG methylation at FANTOM5 enhancers for undifferentiated/stem-like and AC/OC-like IDH-MUT cells (MGH208; $n=123$ cells with matched scRNA-seq and scDNA-seq data). **d**, Proportion of cells with gene expression (RNA read count > 0) and exhibiting above-threshold DNA methylation. Data are shown as mean $\pm$ s.e.m. across all genes with sufficient RNA (expression seen in > 5 cells) and DNA methylation (> 5 CpGs per promoter) information across non-malignant cells ($n=148$) and GBM ($n=765$) and IDH-MUT ($n=670$) malignant cells. $*P < 0.05$. **e**, Left plot, distribution of Spearman's rho for correlation of expression with promoter DNA methylation ($n=1,523$ genes expressed in > 5 cells, DNA methylation at > 5 CpGs per promoter) across GBM ($n=765$) and IDH-MUT ($n=670$) malignant cells. The distribution of Spearman's rho values was compared to the distribution of values obtained with randomly permuted cell labels. Right plot, median values of Spearman's rho for correlation of expression with promoter DNA methylation. See also Extended Data Fig. 8g. **f**, Mean CpG methylation at CTCF-binding sites comparing non-malignant cells ($n=148$) with GBM ($n=765$) and IDH-MUT ($n=670$) malignant cells. **g**, Mean CpG methylation at CTCF-binding sites per quartile of gene expression correlation (Spearman's rho) for previously defined pairs of neighboring genes separated by CTCF-binding sites⁷⁰ in GBM ($n=765$) and IDH-MUT ($n=670$) malignant cells. **h**, The log odds ratio of *PDGFRA-FIP1L1* gene pair coexpression (for both genes, RNA read count > 0) in IDH-MUT and GBM malignant cells. Error bar represents the 95% confidence interval (CI). P values were determined by two-sided Mann-Whitney U test (**a–d,f,g**); by comparing the proportion of cells with gene expression and exhibiting above-threshold DNA methylation in IDH-MUT and GBM cells at each DNA methylation threshold in **d**), two-sided Kolmogorov-Smirnov test (**e**) and Fisher's exact test (**h**). Boxplots represent the median and bottom and upper quartiles; whiskers correspond to 1.5 times the interquartile range.

enhancers were particularly susceptible to hypermethylation in IDH-MUT cells (Mann-Whitney U test, $P < 10^{-16}$; Fig. 3b), which also affected regions enriched for H3K27ac—a histone modification marking active enhancers⁶² (Extended Data Fig. 8a,b). To obtain higher-coverage single-cell DNA methylomes in CpG-sparse regions, such as enhancers, we performed dual-restriction enzyme digestion (HaeIII + MspI) of cells from two IDH-MUT samples (MGH201 and MGH208). This allowed us to increase coverage to a mean of $325,492 \pm 21,118$ unique CpGs per cell as compared to IDH-MUT cells digested with a single restriction enzyme (Extended

Data Fig. 8a), thus enabling more accurate measurement of DNA methylation in regulatory regions. Enhancer hypermethylation was observed in both subsets of cells from IDH-MUT tumors exhibiting the glioma CpG island methylator phenotype (G-CIMP-low and G-CIMP-high subsets) (Extended Data Fig. 8c), supporting the preferential involvement of TET enzymes in the regulation of DNA methylation at enhancers^{59–61}. We observed that enhancer DNA hypermethylation increased with differentiation to AC- and OC-like cells as compared to NPC-like cells (Mann-Whitney U test, $P = 0.016$; Fig. 3c and Extended Data Fig. 8d).

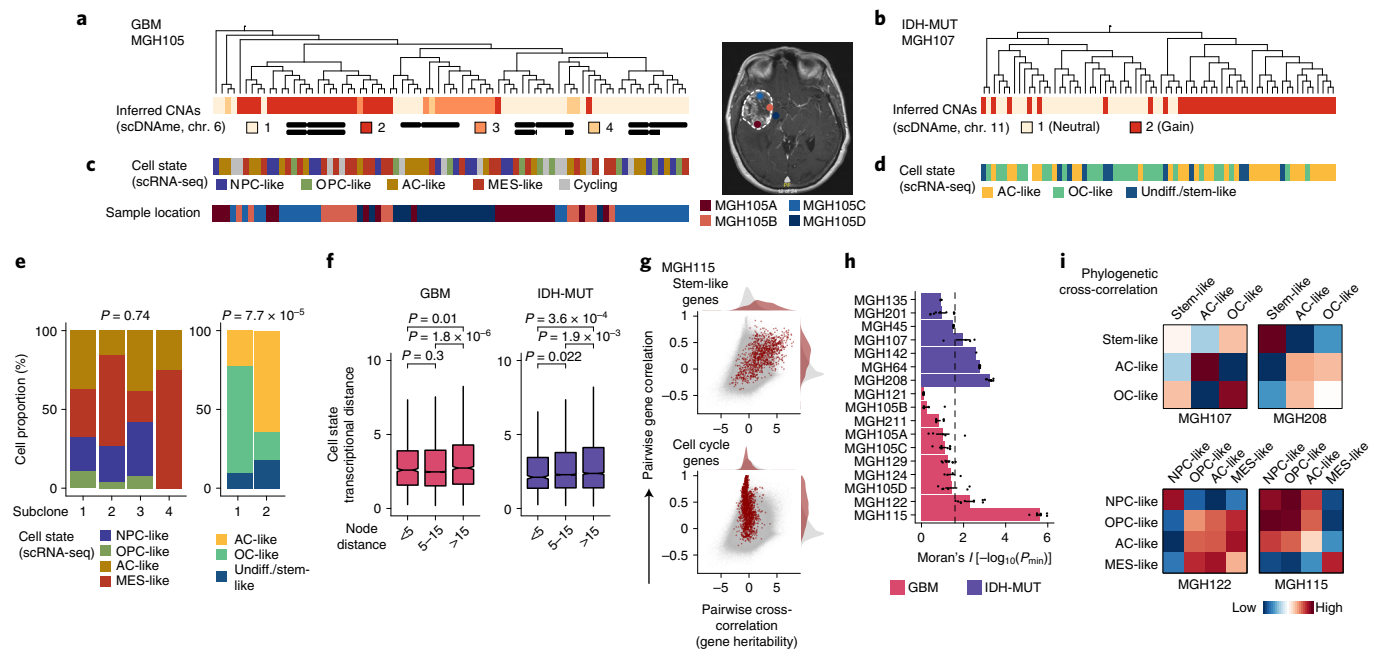


Fig. 4 | Heritability of glioma malignant cell states inferred from lineage tree architectures. **a**, Representative (random cell subsampling within each of the four spatial locations sampled) DNA methylation-based lineage tree of GBM (MGH105) cells, with projection of chromosome 6 inferred CNAs (as defined in Fig. 1f). **b**, Representative DNA methylation-based lineage tree of IDH-MUT (MGH107) cells, with projection of chromosome 11 inferred CNAs. **c**, Projection onto the DNA methylation-based lineage tree of GBM MGH105 (see **a**) of cellular states (top) and sample collection location (bottom). Right, the magnetic resonance imaging (MRI) image from MGH105 indicates the four spatially distinct regions sampled. **d**, Projection onto the DNA methylation-based lineage tree of IDH-MUT MGH107 (see **b**) of cellular states. **e**, Proportion of cells belonging to GBM (MGH105; left) or IDH-MUT (MGH107; right) cellular states in distinct genetic subclones as identified by scDNAme-based CNA inference at chromosome 6 (left) or chromosome 11 (right). **f**, Comparison of transcriptional distances (measured as Euclidean distances between gene module scores) as a function of lineage distance (defined as <5, between 5 and 15, and >15 nodes away) for unique cell pairs from GBM (left; $n = 7$ patients) and IDH-MUT (right; $n = 7$ patients) lineage trees. **g**, Pairwise gene expression correlation (Pearson's) and cross-correlation (heritability). Gray points represent all gene pair relationships, and red points represent gene pair relationships within the same selected gene module (top, stem-like; bottom, cell cycle). Correlation and cross-correlation densities are shown in the plot margins. **h**, Phylogenetic association of cell states on GBM ($n = 7$ patients; $n = 10$ biological replicates) if considering the four spatially distinct regions sampled from MGH105) and IDH-MUT ($n = 7$ patients) lineage trees, as measured by cell state gene module expression autocorrelation with Moran's I (ref. 74). Barplots represent mean $\pm$ s.e.m. Moran's I permutation-based one-sided P values (10^6 permutations) were calculated across lineage tree replicates. The dashed line corresponds to a P value of 0.025. **i**, Heat maps of pairwise cell state phylogenetic associations (gene module cross-correlation analytical z scores). Close phylogenetic associations are shown in warmer colors, and distant associations are shown in cooler colors. P values were determined by two-sided Fisher's exact test (**e**) or two-sided Mann-Whitney U test (**f**). Boxplots represent the median and bottom and upper quartiles; whiskers correspond to 1.5 times the interquartile range.

Cancers are known to exhibit stochastic DNA methylation changes (epimutations), resulting in discordant DNA methylation at neighboring CpGs^{33,63–66}. In line with this notion, single-cell epimutation at promoters was higher overall in malignant cells than in non-malignant cells (Extended Data Fig. 8e). There were more epimutations at promoters in IDH-MUT cells than in GBM cells, in line with a deficiency in TET-mediated demethylation⁵⁸ (Mann-Whitney U test, $P < 10^{-16}$; Extended Data Fig. 8e). This increase in promoter epimutation was associated with decoupling of the typical anticorrelation between gene expression and promoter (transcription start site (TSS) ± 1 kb) DNA methylation⁶⁷ in IDH-MUT malignant cells (Mann-Whitney U test, $P < 0.05$; Fig. 3d and Extended Data Fig. 8f). This decoupling led to a positive correlation between DNA methylation and expression, such that expression of genes central to the oncogenic phenotype (for example, cell cycle and DNA damage response genes) persisted despite high promoter DNA methylation^{68,69} (Fig. 3e and Extended Data Fig. 8g).

An additional mechanism through which hypermethylation in IDH-MUT cells may cause aberrant gene activation is through stochastic hypermethylation of CTCF-binding sites (Mann-Whitney U test, $P < 10^{-16}$; Fig. 3f), with loss of gene insulation

between topologically associating domains (TADs) leading to aberrant enhancer-promoter interactions⁷⁰. To directly assess cell-to-cell variation in CTCF-binding site methylation and insulation efficacy, we identified pairs of neighboring genes separated by TAD-boundary-associated CTCF-binding sites (<180 kb apart (the average contact domain size)⁷⁰) and computed their gene expression correlation as a function of CTCF-binding site DNA methylation. Single-cell CTCF-binding site hypermethylation in IDH-MUT cells correlated with loss of gene insulation (that is, the higher the DNA methylation, the stronger the correlation in the expression of gene pairs across boundaries; Mann-Whitney U test, $P = 1.7 \times 10^{-10}$; Fig. 3g and Extended Data Fig. 8h,i). In line with previous work using bulk sequencing methods⁷⁰, this result suggests that even small changes in DNA methylation are sufficient to disrupt CTCF binding and domain boundaries, thereby affecting gene expression in IDH-MUT gliomas. We further confirmed stronger expression correlation between *PDGFRA*, a prominent glioma oncogene, and *FIP1L1* in IDH-MUT cells than in GBM cells (Fisher's exact test, $P < 10^{-16}$; Fig. 3h), as previously reported⁷⁰. Stochastic methylation of CTCF-binding sites may thus provide the basis for higher transcriptional variation within IDH-MUT tumors by permitting malig-

nant cells to activate alternate gene regulatory programs, eventually leading to the selection of epigenetic clones with higher fitness²⁸.

GBM cells display higher cellular plasticity than IDH-MUT cells. While DNA methylation changes may mark cell states, we and others have previously shown that the large majority of DNA methylation changes in cancer reflect stochastic, passenger events that do not impact gene regulation^{33,63–66,71,72}. These heritable stochastic DNA methylation changes serve as a molecular clock^{33,71–73} and were therefore exploited as native barcodes to infer a high-resolution lineage history of GBM and IDH-MUT cells from primary patient samples (Fig. 4a,b and Extended Data Fig. 9a,b). Projection of information on subclonal CNAs (for example, on chromosome 6 in GBM (MGH105) and chromosome 11 in IDH-MUT glioma (MGH107)) and single-nucleotide variants (SNVs; for example, *RPL5* chr1:g.93303106C>G) onto the lineage trees revealed that genetically defined subclones mapped accurately to distinct clades inferred solely on the basis of DNA methylation information (Fig. 4a,b and Extended Data Fig. 10a; note that chromosomes with CNAs were excluded from DNA methylation tree inference), providing orthogonal validation to lineage tree inference. We further validated that tree topologies were driven primarily by heritable passenger DNA methylation changes by excluding DMRs and PRC2 targets from lineage tree inference (Extended Data Fig. 10c).

In GBM (for example, MGH105), projection of scRNA-seq-derived cell states onto the lineage tree revealed little differential enrichment of the four core cell states in distinct clades of the tree, despite the clades also being marked by CNAs and involving spatially distinct regions of the tumor (Fig. 4c,e and Extended Data Fig. 10a). By contrast, in IDH-MUT samples (for example, MGH107), projection of cellular state onto the lineage trees revealed differential enrichment of the two main differentiated cellular states (AC- and OC-like) in separate clades of the tree, which were also marked by a distinct CNA profile on the long arm of chromosome 11 (11q; Fisher's exact test, $P=7.7 \times 10^{-5}$; Fig. 4d,e and Extended Data Fig. 10b). These observations may suggest a model of higher cellular plasticity in GBM while there is a more stable differentiation hierarchy in IDH-MUT tumors¹⁶ and raise the question of the extent to which glioma cell states are heritable.

To investigate the heritability of glioma cell states, we assessed phylogenetic association of cellular states on the lineage tree as a proxy for the heritability of gene expression programs. We observed decreased transcriptional similarity between glioma cells as a function of their lineage distance (Fig. 4f and Extended Data Fig. 10d,e). We also compared transcriptional correlation to phylogenetic cross-correlation⁷⁴ for pairs of genes. As expected, genes within the same module (for example, cell cycle or stem-like genes) exhibited highly correlated transcription. However, stem-like genes (expressed in NPC- and OPC-like cells) tended to also have high phylogenetic cross-correlation, reflecting heritable expression of these lineage-specific genes over the course of cellular divisions. By contrast, cell cycle genes, despite exhibiting highly correlated expression, did not show high phylogenetic cross-correlation, reflecting their transient, non-heritable status (Fig. 4g and Extended Data Fig. 10f). To directly assess cell state heritability, we measured with Moran's *I* (ref. ⁷⁴) the autocorrelation between cell state gene module expression and found that the majority of IDH-MUT samples (4 of 7) and a subset of GBM samples (2 of 7) showed significant cell state heritability (Fig. 4h, Extended Data Fig. 10g,h and Supplementary Table 9).

Focusing on glioma samples with the highest degree of cell state heritability, we observed that cell state lineage proximity mirrored transcriptional similarity; in GBM, NPC- and OPC-like cells tended to cluster together on the lineage trees, and AC-like cells exhibited the closest phylogenetic proximity to MES-like cells. This pattern of phylogenetic cross-correlation may indicate that cell state heritability

dynamics in GBM cohere with neurodevelopment trajectories (Fig. 4i, Extended Data Fig. 10i and Supplementary Table 10). In IDH-MUT tumors, this analysis revealed two distinct clusters of differentiated cell states in the majority of patients. This result likely reflects the branched unidirectional developmental hierarchy, with activation of neural stem-cell programs at the top of the hierarchy that branches into two distinct cellular states resembling astrocytic and oligodendrocytic lineages⁷ (Fig. 4i and Extended Data Fig. 10i).

These heritability findings prompted us to quantify the transition dynamics governing the distribution of glioma cell states across lineage trees. We hypothesized that plastic differentiation hierarchies (that is, those with a high degree of dedifferentiation in which differentiated cells can more easily revert to stemness) would result in lineage trees where the cell states were distributed more randomly across clades, whereas a strict unidirectional hierarchy would result in lineage trees with cell states that were more clustered, as observed in GBM and IDH-MUT tumors, respectively. In line with this hypothesis, simulated lineage trees with varying rates of dedifferentiation in comparison to stem-like cell self-renewal showed that the phylogenetic clustering of cell states (as measured by Moran's *I*) decreased as the rate of dedifferentiation increased (Fig. 5a).

To examine this hypothesis directly in patient samples, we inferred cell state growth and transition rates from glioma phylogenetic trees with leaves annotated for cell state. Specifically, we adapted a maximum-likelihood method of binary character evolution and speciation from comparative phylogenetics^{75,76} (Methods, Extended Data Fig. 10j,k and Supplementary Table 11). To validate the model's parameter estimates, we used two sources of orthogonal data. First, we compared the model's estimates of growth in differentiated-like versus stem-like states to cycling rates derived from the expression profiles and observed high correlation (Spearman's $\rho=0.8$, $P=0.014$; Fig. 5b). Second, we found that the model's estimates of dedifferentiation correlated with dedifferentiation rates inferred from RNA velocity estimation³¹ of gene module trajectories (Spearman's $\rho=0.71$, $P=0.0014$; Fig. 5c). We further validated the model's estimates by excluding DMRs and PRC2 targets from lineage tree inferences (Extended Data Fig. 10l), confirming again that DNA methylation-derived tree topology reflects stochastic passenger DNA methylation changes rather than cell state encoding.

When the binary character evolution method was applied to IDH-MUT samples, the model predicted a low rate of dedifferentiation in comparison to stem-like cell self-renewal (Fig. 5d and Extended Data Fig. 10m), in line with the highly structured lineage trees for these tumors (Fig. 5a,e). By contrast, GBM samples showed a significantly higher level of dedifferentiation (Mann–Whitney *U* test, $P=0.0046$; Fig. 5d and Extended Data Fig. 10m), in line with the lower degree of cell state clustering on the trees and lower transcriptional similarity by lineage distance (Fig. 5a,e). Together, these data demonstrate that cell states are heritable across malignant gliomas. However, while in IDH-MUT tumors, differentiation far outpaces dedifferentiation in line with a standard hierarchical model⁷, GBM tumors harbor a higher degree of cell state plasticity allowing replenishment of the ranks of stem-like cells through dedifferentiation (Fig. 5f).

Discussion

Studies across cancer types have shown that heterogeneous transcriptional cell states within a single tumor contribute to tumor initiation and progression^{1,4–7}. In glioma, cellular state diversity mirrors neurodevelopmental trajectories^{7,17–23}. Here, through the application of multiomics single-cell sequencing to primary glioma clinical samples, we provide evidence that DNA methylation changes reflect glioma cellular states and may contribute to their propagation.

Specifically, we showed that IDH-MUT cells exhibit preferential enhancer hypermethylation with cell differentiation. Enhancers,

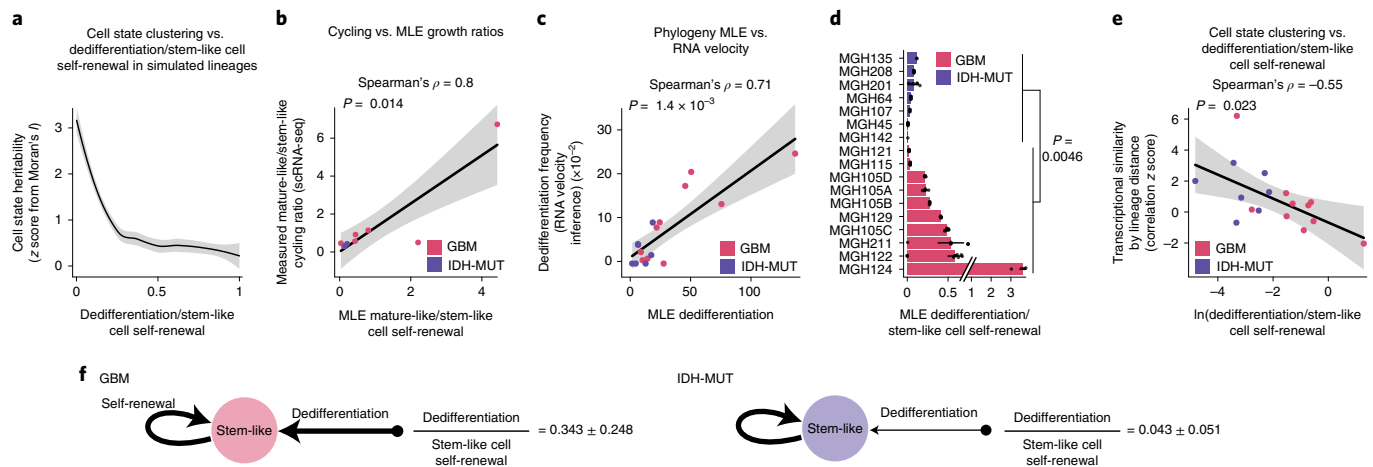


Fig. 5 | GBMs exhibit higher cellular plasticity while IDH-MUT gliomas have a more stable differentiation hierarchy. **a**, Simulated lineage trees ($n=1,000$) with varying rates of dedifferentiation compared to stem-like cell self-renewal (x axis) as a function of phylogenetic association of cellular states (as measured by z scores from Moran's I ; y axis). The LOESS regression line (black) with its 95% confidence interval (gray) is shown. **b**, Comparison of the mathematical model's estimates (MLE, maximum-likelihood estimation; weighted median across lineage tree replicates for each GBM (pink) and IDH-MUT (purple) sample) of cell state self-renewal in differentiated-like versus stem-like states with cycling rates derived from the scRNA-seq expression profiles. Spearman's rho is indicated. Only samples with at least two cycling stem-like and two cycling differentiated-like cells were used (Methods). The linear regression line (black) with its 95% confidence interval (gray) is shown. **c**, Same as in **b** for comparison of the mathematical model's estimates of dedifferentiation with dedifferentiation rates provided by RNA velocity estimation³¹ (Methods). The linear regression line (black) with its 95% confidence interval (gray) is shown. **d**, Rates of dedifferentiation compared to stem-like cell self-renewal (as estimated by mathematical modeling) in GBM ($n=7$ patients; $n=10$ if considering the four spatially distinct regions sampled from MGH105) and IDH-MUT ($n=7$ patients) tumors across lineage tree replicates (Methods). Barplots represent median $\pm$ median absolute deviation (MAD) across lineage tree replicates. Medians are weighted to balance plates with a disparate number of lineage tree replicates within samples. **e**, Dedifferentiation/stem-like cell self-renewal ratio (weighted median across lineage tree replicates for each GBM (pink) and IDH-MUT (purple) sample; x axis) compared to cell state clustering on the lineage tree as measured by transcriptional similarity (mean across tree replicates of a gene module by lineage distance Pearson's correlation z score; 1,000 permutations). The linear regression line (black) with its 95% confidence interval (gray) is shown. **f**, Data-driven model of cell state transition dynamics inferred from DNA methylation-based lineage trees. The median $\pm$ MAD dedifferentiation/stem-like cell self-renewal ratios across GBM ($n=7$ patients) and IDH-MUT ($n=7$ patients) samples are shown. P values were determined by two-sided Mann-Whitney U test (**d**), comparing the weighted median dedifferentiation/stem-like cell self-renewal ratio of GBM samples with the weighted median dedifferentiation/stem-like cell self-renewal ratio of IDH-MUT samples.

owing to their lower transcription factor occupancy as compared to promoters⁷⁷, may be less resistant to DNMTs and thus more prone to hypermethylation, which is canonically balanced by the action of TET enzymes in physiological contexts⁶¹. In IDH-MUT malignant cells, defects in TET-mediated demethylation caused by 2HG may thus lead to preferential enhancer hypermethylation⁶⁰. In addition, enhancers have been shown to exhibit highly dynamic DNA methylation during differentiation^{78–80}, in line with our data showing increased enhancer DNA methylation with glioma differentiation. While the relatively modest magnitude of DNA methylation changes observed in our study may be partly due to the sparsity of the single-cell RRBS data, our work also suggests that small increases in DNA methylation in otherwise typically unmethylated regions are sufficient to impact gene expression and can be associated with gene silencing (Fig. 3d), as previously reported across cancer types^{33,63,81–84}, including in glioma⁸⁵. Indeed, our multimodality sequencing technology that couples single-cell DNA methylomes with whole-transcriptome sequencing allowed the exploration of methylation–transcription relationships at the single-cell level, revealing that aberrant epigenetic patterning is at play in IDH-MUT gliomas. This included decoupling of promoter methylation–expression relationships, whereby expression of genes central to the oncogenic IDH-MUT phenotype persists despite high promoter DNA methylation, as well as disruption of CTCF-mediated insulation.

In GBM, direct comparison of epigenetic profiles across cell states suggests that the interaction between DNA methylation and PRC2 is an important contributor to GBM cell differentiation. The main role of PRC2 is to catalyze H3K27me3 deposition to repress

lineage-specific developmental genes in both normal and neoplastic stem cells^{86,87}. At these genes, H3K27me3 is largely enriched at promoters along with H3K4me3, an activating histone mark⁸⁸. These bivalent poised promoters in stem cells largely resolve to either an active (H3K4me3-only) or repressed (H3K27me3-only) state during differentiation. While PRC2 target hypermethylation has previously been extensively reported in cancer⁵², we observed that stem-like GBM cells are protected from this phenomenon, likely owing to PRC2 binding protecting these sites from DNA methylation, in line with data from neurodevelopment^{89,90}. This may also underlie the enhanced chromatin accessibility signal that we observed at hypomethylated PRC2 targets in GBM stem-like cells⁹¹. By contrast, differentiated-like GBM cells may reinforce gene silencing by increasing the length of H3K27me3 domains or through complementary silencing mechanisms involving DNA methylation^{87,92}. In line with this model, we observed more than twofold-higher expression of PRC2 targets in stem-like cell states in comparison to more robust silencing involving DNA methylation in more differentiated cell states. Thus, our multimodal single-cell analyses support a critical role for PRC2 in maintaining GBM cellular states, suggesting a model in which PRC2 targets are maintained in a hypomethylated state in glioma stem-like cells, allowing their reactivation in response to stimuli, thereby ultimately providing a key mechanism for stemness maintenance⁹⁰ and tumor progression⁹³.

The observed parallels between glioma differentiation and neural development invoke the question of whether gliomas follow unidirectional differentiation hierarchies or more reversible bidirectional cell state transitions^{21,55}. As we seek to therapeutically target defined

glioma cell states, such as stem-like cells⁹⁴, it is critical to dissect the relative rates at which other cells revert to assume the role of stem cells. To address this question, we integrated lineage histories derived from heritable stochastic DNA methylation changes with scRNA-seq-derived cell states in single-cell multiomics data. We demonstrated that in IDH-MUT glioma differentiation far outpaces dedifferentiation, in line with a model in which stem-like cells are self-renewing and reside at the apex of the cellular hierarchy⁷. By contrast, in GBM, cells demonstrated the capacity to dedifferentiate into stem-like states, providing evidence for plastic bidirectional cell state transitions, as also observed in other cancer types^{95,96}. Such plastic differentiation topologies may result from relaxation of epigenetic identity barriers^{28,63,80} and in turn may empower positive selection⁹⁷ to enhance the evolutionary capacity of gliomas.

Our work has several limitations. The MscRRBS platform only captures approximately 10% of the targeted methylome for a single cell owing to the sparsity of single-cell data³³. We have thus implemented several analytical approaches to mitigate the sparsity of the single-cell methylomes, including averaging DNA methylation levels across defined genomic windows and regions or aggregating DNA methylation signal over multiple single cells within a sample. We further note that, while DNA methylation is one of the central mechanisms for propagating stable epigenetic information across cell division⁵⁴ and accumulating data suggest that malignant cell states are propagated epigenetically^{4,27–29}, the nature of the causal relationship between DNA methylation and the establishment of stable cellular identity is still under debate⁹⁸. Nonetheless, we envision that future advances in both experimental technologies and data analysis methods⁹⁹ will enable more accurate measurement of DNA methylation across the genome in single cells, as well as a better understanding of the causal relationship between DNA methylation and transcriptional cell states.

In conclusion, cell state diversity and tumor evolution are often studied independently. The data presented herein show that single-cell multiomics analysis of clinical samples can help draw together these disparate frameworks, through the unique lens of a high-resolution phylogenetic tree coupled with leaf annotation for current phenotypic states. This new perspective allows transcriptional cell state diversity to be connected with fundamental evolutionary properties such as heritability and cell state transition dynamics, opening up new horizons for the study of human somatic evolution in both malignant and healthy tissues.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41588-021-00927-7>.

Received: 9 September 2020; Accepted: 30 July 2021;

Published online: 30 September 2021

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Methods

Study participants. Adult patients included in this work provided preoperative informed consent to take part in the study according to institutional review board protocol Dana-Farber/Harvard Cancer Center 10-417. Patients were male and female. Clinical characteristics are summarized in Supplementary Table 2.

Tumor acquisition and single-cell sorting. Fresh tumor specimens were collected on PBS (Gibco) and mechanically dissociated into small pieces of 0.5–1 mm with a disposable sterile scalpel. They were further dissociated into single-cell suspensions using the enzymatic brain tumor dissociation kit (P) from Miltenyi Biotec, following the manufacturer's protocol. Viable single cells were sorted into individual wells of a 96-well twin.tec PCR plate (Eppendorf) that contained 10 μ l per well of TCL buffer (Qiagen) with 1% β -mercaptoethanol (see the Supplementary Note and Supplementary Fig. 1 for details). Plates were frozen on dry ice immediately after sorting and stored at -80°C before joint MscRRBS and whole-transcriptome library preparation and sequencing.

Joint MscRRBS and scRNA-seq library construction. MscRRBS and whole-transcriptome library preparation and sequencing were performed as previously described³³ (see the Supplementary Note for details). To obtain higher-coverage single-cell DNA methylomes, dual-restriction enzyme digestion of cells from two IDH-MUT samples (MGH201 and MGH208) was performed. This allowed us to increase coverage to $325,492 \pm 21,118$ unique CpGs per cell (~ 2 -fold increase) as compared to IDH-MUT cells digested with a single restriction enzyme, thus enabling more accurate measurement of DNA methylation in regions that are captured less efficiently with standard RRBS, such as enhancers and CTCF-binding sites (Extended Data Fig. 8a).

MscRRBS read alignment. Each pool of 96 cells was first demultiplexed by Illumina 17 barcodes (Supplementary Table 1), resulting in four pools of 24 cells. Each pool of 24 cells was further demultiplexed by unique cell barcodes (Supplementary Table 1). Quality control, trimming and alignment of MscRRBS data were then performed³³ (see the Supplementary Note for details). Cells with coverage of at least 50,000 unique CpGs and a bisulfite conversion rate of at least 99% were retained for downstream analyses (Supplementary Tables 2 and 3).

scRNA-seq and differential gene expression analysis. Sequenced read fragments were mapped against the GRCh38 (hg38 Ensembl version 94) genome assembly using the 2pass default mode of STAR¹⁰⁰ (v2.5.2a). The number of read counts overlapping annotated genes was determined using RSEM¹⁰¹ v1.3.1 (rsem-calculate-expression). Cells with mitochondrial and ribosomal read counts of less than 20% and a minimum of 2,000 detected genes were retained for downstream analyses (Supplementary Tables 2 and 3). Differential gene expression analyses were performed using a negative binomial model with observational weights to account for zero inflation¹⁰². Specifically, we used ZINB-WaVE¹⁰³ (v1.6) to estimate a set of observational weights and edgeR (v3.26.8) to test for differential expression using a weighted F -statistic approach¹⁰⁴. We defined differentially expressed genes by adjusting nominal P values using a BH FDR procedure (cutoff of adjusted P value < 0.05), with an additional criterion of an absolute $\log_2$ (fold change) value of > 1 (Extended Data Figs. 4b and 5f).

Identification of non-malignant cell types. To classify all cells passing scRNA-seq quality control (Supplementary Table 3) into malignant or non-malignant cells (Fig. 1b), we normalized gene count matrices, performed dimensionality reduction and corrected for patient batch effects using the ZINB-WaVE method¹⁰³ (v1.6; parameters: $K = 30$, $X = \text{"patient sample"}$). To classify all cells passing scDNA-seq quality control (Supplementary Table 3) into malignant or non-malignant cells (Fig. 1b), we focused on 1,300 CpG sites that were identified as glioma related by a previous TCGA bulk DNA methylation study³⁹. We generated a window of 1,000 bp around each CpG (resulting in 996 windows) and averaged the DNA methylation within each window. We then imputed the missing values in the windows using KNN with $N = 5$. We used the scanpy package¹⁰⁵ (v1.4.4) to cluster cells. For visualization, we generated a UMAP cell embedding using the umap function (v0.2.3.1) with default settings.

Single-cell differential methylation analysis. For each cell, Bismark methylation extractor output files (containing information on the methylation state of each individual CpG) were intersected with different genomic regions investigated (for example, promoters and enhancers) using BEDTools¹⁰⁶ (v2.27.1). A generalized linear model was then built to predict the DNA methylation for a given genomic region between groups of cells on the basis of transcriptionally defined malignant cellular states (see the Supplementary Note for details). We defined regions with a Student's t -test P value < 0.05 and an absolute DNA methylation difference of $\geq 5\%$ as differentially methylated to nominate candidate genes for subsequent gene set enrichment analysis.

CNA inference from single-cell DNA methylation data. To estimate CNAs using scDNA-seq data, we first split the genome of each cell into windows of equal length (20 Mb) and obtained the number of CpGs per window with a sliding window of 5 Mb. We subsequently normalized the number of CpGs per window by the

total number of CpGs for each cell. Cells classified to each of the non-malignant cell types (see "Identification of non-malignant cell types") were used to define a baseline normal karyotype. We then divided the number of normalized CpGs per window in each malignant cell by the median normalized number of CpGs in the set of non-malignant cells. The resulting copy number estimates were log₂ transformed. Missing values were replaced by the value zero (Extended Data Figs. 1a and 2a). For CNA analysis of the *EGFR* locus, we applied the above-described approach using a 0.1-Mb window (with a sliding window of 0.02 Mb) centered on the *EGFR* locus on chromosome 7 (Fig. 1e and Extended Data Fig. 3d,f). We further localized the start and end points of aberrant copy number regions of the pseudo-bulk averages (mean of CNAs across individual malignant cells) using the circular binary segmentation algorithm implemented in the R package DNACopy¹⁰⁷ (v1.60.0). See the Supplementary Note for further details.

Glioma DNA methylation subtype (LGm1–LGm6) single-cell projection. To bridge the 450K methylation array and MscRRBS technologies, we created a window of 1,000 bp around each 450K probe obtained for 932 glioma samples from TCGA^{40,41}, averaging the DNA methylation within each window (450K probes for the TCGA samples and single CpGs for MscRRBS), resulting in 996 windows. We further filtered the data by retaining (1) 450K probes that were detected in at least 20 bulk TCGA samples; (2) single cells with at least 50,000 detected CpG sites; (3) windows containing more than 5 CpGs per cell; and (4) windows for which more than 10 single cells had at least 1 CpG in them. After filtering, we retained 979 windows and imputed missing values in the windows using KNN with $N = 5$. We then trained a logistic regression multiclass classifier on the 932 TCGA glioma samples, achieving 0.94 accuracy, and applied it to pseudo-bulk DNA methylation profiles for malignant cells in our samples (Fig. 1h) to assign each glioma single cell to one of the six bulk DNA methylation subtypes (Fig. 1i).

Definition of single-cell gene signature scores. Single-cell gene signature scores were defined as previously described^{17,21} (see the Supplementary Note for details).

Assignment of glioma cells to expression cell states. We classified glioma cells by expression cell state on the basis of gene modules and cell cycle programs as previously described^{7,21} (see the Supplementary Note for details).

Analysis of TCGA patient samples. To examine the association between GBM stem-like states and PRC2 target activity in a larger sample cohort, we leveraged 67 GBM samples from the TCGA collection with matched bulk RNA-seq and 450K methylation profiles (Fig. 2i,j and Extended Data Fig. 5j). We computed differentiation scores (defined as the difference in gene module scores between AC/MES-like and NPC/OPC-like cellular states) where the gene signatures for each of the four states were taken from the previously described gene module signatures²¹. We calculated the mean DNA methylation at PRC2 target promoters by averaging the DNA methylation for the 450K probes mapping within PRC2 target genes⁴⁶.

Chromatin state analysis. To explore the link between differentially methylated promoters (see "Single-cell differential methylation analysis") and histone marks, we interrogated differentially methylated promoters for enrichment of histone marks with non-overlapping regulatory functions (H3K4me3, H3K27ac, H3K4me1, H3K36me3 and H3K27me3) using previously published ChIP-seq maps⁴⁷ of GBM cancer stem cells ($n = 4$ lines derived from different human gliomas (MGG23CSC, MGG4CSC, MGG6CSC and MGG8CSC)). In Extended Data Figs. 5c–e and 7k,l, chromatin states across the genome were defined using ChromHMM¹⁰⁸ (v1.20), which is based on a multivariate hidden Markov model (HMM), using H3K4me3, H3K27ac, H3K27me3, H3K36me3 and H3K4me1 from the above-described previously published datasets⁴⁷ as input (the MGG8CSC sample was used as it was the only one where all five main histone marks were profiled). See the Supplementary Note for further details.

Single-cell DNA methylation–gene expression correlation analysis. Single-cell DNA methylation–gene expression correlation analysis was performed as previously described³³ (see the Supplementary Note for details).

Lineage tree inference. We generated DNA methylation-based lineage trees by applying a tree searching maximum-likelihood algorithm based on binary DNA methylation values as previously described³³ (see the Supplementary Note for details). Projection of information on subclonal CNAs (for example, on chromosome 6 in GBM (MGH105) and chromosome 11 in IDH-MUT glioma (MGH107)) and SNVs (for example, in *RPL5*) onto the lineage trees revealed that genetically defined subclones mapped accurately to distinct clades inferred solely on the basis of DNA methylation information, providing orthogonal validation to lineage tree inferences (Fig. 4a,b and Extended Data Fig. 10a). We further validated that lineage tree topologies were driven by heritable stochastic passenger DNA methylation changes by excluding CpGs belonging to DMRs and PRC2 targets from lineage tree inference (Extended Data Fig. 10c). To compare inferred lineage trees, we computed the pairwise Robinson–Foulds (RF) distance—a measure of tree structure similarity between two given trees¹⁰⁹. RF distances were normalized by the total number of internal edges in respective pairs of trees (normalized RF distance).

Phylogenetic association. To quantify the association of different cell states and transcriptional patterns on the DNA methylation-based lineage trees, we used Moran's I (ref. ⁷⁴), a classic measure of spatial association (that is, autocorrelation) used to detect phylogenetic signal¹¹⁰, as well as its multivariate generalization, a measure of spatial cross-correlation^{111,112}. Conceptually, Moran's I is a weighted correlation metric, as its calculation is similar to that of Pearson's correlation coefficient but with measurements weighted by proximity. To compute Moran's I for an n -cell lineage tree, we first organize single-cell measurements into a column-standardized matrix X (centered with mean 0 and population standard deviation of 1), consisting of n rows corresponding to cells and m columns corresponding to single-cell measurements. Then, the data matrix X and its transpose (notated with superscript T) is right and left multiplied with the proximity matrix W ,

$$I = X^T W X.$$

Each element of the $n \times n$ proximity matrix W_{ij} records the inverse node distance between cells i and j , with diagonal elements set to 0, and normalized such that $\sum_{ij} W_{ij} = 1$. Measurements contained in matrix X could correspond to gene expression (as in Fig. 4g), gene module scores (for example, in Fig. 4h) or cell states (for example, in Fig. 5a). When $m = 1$, this metric becomes the classic univariate Moran's I . When $m > 1$, each element of the $m \times m$ matrix I_{yx} measures the phylogenetic cross-correlation between measurements y and x . High values within I indicate phylogenetic co-clustering, whereas low values indicate phylogenetic dispersion.

To assess the heritability of glioma cell states, we measured the phylogenetic autocorrelation of each cell state gene module (using univariate Moran's I) and assessed significance with a one-sided permutation test (with 10^6 leaf permutations) for each tree replicate. To improve resolution, we first recomputed GBM module scores, pooling the NPC1-like and NPC2-like gene sets and the MES1-like and MES2-like gene sets, and removed cells without matching scRNA-seq information. As both GBM and IDH-MUT samples contained multiple cell states at different frequencies, to summarize a sample's transcriptional heritability, we used the most heritable gene module for each tree, as represented by its permutation test $-\log_{10}(P \text{ value})$. As we had multiple lineage tree replicates per sample plate, we arrived at a plate heritability score by averaging the tree replicate $-\log_{10}(P \text{ values})$. For patient samples with multiple plates, scores for only the least variable plate (measured by RF distance; see "Lineage tree inference") are shown (Fig. 4h). Heritability scores for all plates are shown in Extended Data Fig. 10g and included in Supplementary Table 9.

To further understand how cell states were co-distributed/dispersed across lineage trees, we also measured gene module cross-correlation (multivariate Moran's I). Cross-correlations for each tree replicate were transformed into z scores using moments of the statistic derived by Czaplewski and Reich¹¹² and were then averaged for each sample. Analytical z scores were used to increase computational efficiency and closely matched leaf-permutation-based z score estimates. Moran's I z score heat maps for representative lineages are shown in Fig. 4i. These heat maps illustrate which cell states form clusters and how pairs of different cell states cluster together on lineage trees. Close and distant phylogenetic associations are shown in warmer and cooler colors, respectively.

Finally, to study the phylogenetic distribution of transcription at the single-gene level, we compared cross-correlations and correlations for all available (2,000 most variable genes selected with Seurat), stem-like (that is, NPC-like and OPC-like) and cell cycle genes in glioma samples with high gene module transcriptional heritability (MGH115 and MGH122) (Fig. 4g and Extended Data Fig. 10f). For each available gene, all pairwise Pearson's correlations and cross-correlations (mean tree replicate analytical z scores) were plotted, with self-correlations and autocorrelations omitted. Densities are shown for all gene pairs (gray) and for genes from the selected module (red) in plot margins.

Mathematical model of glioma evolutionary dynamics. To model glioma evolutionary dynamics, we adapted a mathematical model of binary state speciation and extinction (BiSSE) from comparative phylogenetics⁷⁵. The BiSSE approach models speciation, extinction and character transition rates as a dynamical system, where species in character state k (either 0 or 1) speciate at rate λ_k . Species transition from state 0 (1) to state 1 (0) at rate q_{01} (q_{10}). This mathematical framework can be translated to tumor dynamics, where λ and q measure cell-state-specific growth (self-renewal) and transition (that is, differentiation and dedifferentiation) rates. In this application, we set the binary character trait to be the tumor cell state, either stem-like ($k=0$) or mature-like ($k=1$). As we are interested in net cell state growth rates, we use a Yule (pure birth) version of the model. The change in the number of cells $n_k(t)$ in state k at time t is described by the following dynamical system (Extended Data Fig. 10j,k):

$$\frac{d}{dt}n_0(t) = (\lambda_0 - q_{01})n_0(t) + q_{10}n_1(t)$$

$$\frac{d}{dt}n_1(t) = (\lambda_1 - q_{10})n_1(t) + q_{01}n_0(t).$$

To apply this method to samples from patients with GBM, we binned cells with a maximum gene module score of NPC- or OPC-like as 'stem-like' and those with a score of AC- or MES-like as 'mature-like'. For IDH-MUT samples, cells with a maximum gene module score of AC- or OC-like were binned to the mature-like cell state and cells with a maximum module score of stem-like remained classified as stem-like. Before assigning cell states, cells without scRNA-seq data were removed and gene module scores for GBM samples were recomputed by first pooling NPC1/NPC2-like and MES1/MES2-like genes into one module each.

Maximum-likelihood estimation of evolutionary dynamics. To infer the tumor growth and transition rates that generated the observed phylogenies, we used maximum-likelihood estimation. We generated a likelihood function using `make.bisse()` from `diversitree` v0.9.15 (ref. ¹¹³), using a Yule version of the model and a sampling fraction of 10^{-6} , as our lineages represent a tumor sampling. We assumed that the root of each tree was in the stem-like state and otherwise used default settings. As the BiSSE method requires ultrametric trees, we converted our trees using `force.ultrametric()` with the 'extend' method in the R package `phytools` (v0.7.70)¹¹⁴.

To minimize the chances of reaching a local, instead of a global, maximum estimate, we initiated the maximum-likelihood searches from 100 randomly generated starting points (initial BiSSE parameter values) using simulated annealing with the R package `GenSA` (v1.1.7)¹¹⁵, searching parameter values bounded by 10^{-4} and 500, and allowed for a maximum of 1,000 iterations with a stop threshold of 10^{-8} . After these initial searches, we used `mle2()` from the `bbmle` R package (v1.0.23.1)¹¹⁶, initializing each maximum-likelihood search with a simulated annealing estimate, using the L-BFGS-B optimization method, lower and upper bounds of 10^{-4} and 500, and a maximum of 1,000 iterations.

After 100 maximum-likelihood searches per tree replicate, the BiSSE parameter scheme with the highest likelihood among the individual runs that converged without error was selected. To arrive at a final parameter set estimate for each biological sample, we used the weighted median of the maximum-likelihood estimates (Supplementary Table 11), weighting each plate for a sample equally. Outlier tree replicates estimated from the same cells with an estimated dedifferentiation/stem-like cell self-renewal (q_{10}/λ_0) ratio greater than 5 MAD above the median were removed. Maximum-likelihood estimates for GBM and IDH-MUT dynamics (Extended Data Fig. 10j) represent the median of patient-sample-weighted median estimates. The ratio of median replicate estimates of q_{10}/λ_0 was significantly larger in GBM than in IDH-MUT samples (Fig. 5f). For patient samples with multiple plates, q_{10}/λ_0 for only the least variable plate (measured by RF distance; see "Lineage tree inference") is shown (Fig. 5d), and the weighted median of q_{10}/λ_0 for all plates is shown in Extended Data Fig. 10m and included in Supplementary Table 11. The P value in Fig. 5d was calculated by Mann-Whitney U test by comparing the weighted median of q_{10}/λ_0 between GBM and IDH-MUT samples using all plates. Lastly, we validated the maximum-likelihood estimates by excluding DMRs and PRC2 targets from lineage tree inference (Extended Data Fig. 10l), confirming that DNA methylation-derived tree topology reflects stochastic passenger DNA methylation changes rather than marking cell states.

Statistical methods. Statistical analysis was performed with Python 3.0 and R version 3.6.1. Categorical variables were compared using the hypergeometric test or Fisher's exact test. Continuous variables were compared using the Mann-Whitney U test, Student's t test, nonparametric permutation test or Kolmogorov-Smirnov test, as appropriate. P values were adjusted for multiple comparisons using the BH FDR adjustment procedure. All P values are two sided and were considered significant at the 0.05 level unless otherwise noted.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Processed data generated for this study are available through the NCBI Gene Expression Omnibus (GEO) under accession number [GSE151506](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE151506). Raw data access can be requested through the Data Use Oversight System (DUOS) Dataset Catalog with dataset ID DUOS-000133 as well as the European Genome-phenome Archive (EGA) with dataset ID [EGAS00001005472](https://www.ebi.ac.uk/ega/dataset/EGAS00001005472). The data can be visualized and interrogated through the Broad Institute's Single-Cell Portal at https://singlecell.broadinstitute.org/single_cell/study/SCP936. scATAC-seq data are available at the EGA repository under [EGAS00001002185](https://www.ebi.ac.uk/ega/dataset/EGAS00001002185), [EGAS00001001900](https://www.ebi.ac.uk/ega/dataset/EGAS00001001900) and [EGAS00001003845](https://www.ebi.ac.uk/ega/dataset/EGAS00001003845) and at NCBI GEO under accession number [GSE138794](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE138794). TCGA data (DNA methylation, gene expression and clinical profiles) are available from the TCGA database (<https://cancergenome.nih.gov/>). ChIP-seq data are available at NCBI GEO under accession number [GSE46016](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE46016).

Code availability

The analytic code used for this work is provided for noncommercial use at <https://doi.org/10.5281/zenodo.4776456> (ref. ¹¹⁷).

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Acknowledgements

We thank members of the Landau and Suvà laboratories for constructive discussions, the Epigenomics Core Facility at Weill Cornell Medical College for technical help and E. Rheinbay at the Massachusetts General Hospital Cancer Center for whole-exome sequencing data processing. This project and R.C. have received funding from the European Union's Horizon 2020 research and innovation program under Marie

Skłodowska-Curie grant agreement no. 750345. F.G. was supported by NIH K99/R00 Pathway to Independence Award (NCI K99CA248955). D.S. was supported by EMBO long-term fellowship ALTF (570-2017) and by the Schmidt Family Foundation. J.K. was supported by an HFSP long-term fellowship (LT000452/2019-L). A.R. was supported by funds from the Howard Hughes Medical Institute, the Klarman Cell Observatory, the STARR Cancer Consortium, NCI grant 1U24CA180922, NCI grant R33CA202820, Koch Institute support (core) grant P30CA14051 from the NCI, the Ludwig Center and the Broad Institute. L.N.G.C. was supported by NIH award K12CA090354. This work was supported by grants to M.L.S. from the Mark Foundation (Emerging Leader Award), the Sontag Foundation (Distinguished Scientist Award), the MGH Research Scholars, and NCI R37CA245523 and NCI R01CA258763 (to M.L.S. and D.A.L.). D.A.L. was supported by the Burroughs Wellcome Fund Career Award for Medical Scientists, the Pershing Square Sohn Prize for Young Investigators in Cancer Research, the NIH Director's New Innovator Award (DP2-CA239065), the Sontag Foundation (Distinguished Scientist Award, SFI 203261-01), the William Rhodes and Louise Tilzer-Rhodes Center for Glioblastoma at NewYork-Presbyterian Hospital (NYPH 203205-01) and NHGRI RM1HG011014-01. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Author contributions

R.C., F.G., D.S., J.S.S., M.L.S. and D.A.L. conceived the project, designed the study and interpreted results. R.C., H.R.W., A.R.R., C.S., A.A. and J.P. performed patient selection, collected primary glioma single cells and generated single-cell sequencing data. F.G., D.S., J.S.S., L.K., S.D.D. and F.I. performed computational analyses. S.G., L.N.G.C., J.K., T.B., C.M., O.R.-R., A.R., M.L.S. and D.A.L. provided experimental and analytical support. F.G., R.C., D.S., J.S.S., M.L.S. and D.A.L. wrote the manuscript with comments and contributions from all authors.

Competing interests

M.L.S. is an equity holder, scientific cofounder and advisory board member of Immunitas Therapeutics. A.R. is a founder and equity holder of Celsius Therapeutics, is an equity holder in Immunitas Therapeutics and, until 31 July 2020, was a scientific advisory board member of Syros Pharmaceuticals, Neogene Therapeutics, Asimov and ThermoFisher Scientific. Since 1 August 2020, A.R. has been an employee of Genentech. Since 19 October 2020, O.R.-R. has been an employee of Genentech. D.A.L. is an equity holder, scientific cofounder and advisory board member of C2i Genomics and a scientific advisory board member for Mission Bio. The authors declare that these activities are not related to the research reported in this publication and have not influenced the conclusions in this manuscript. The remaining authors declare no competing interests.

Additional information

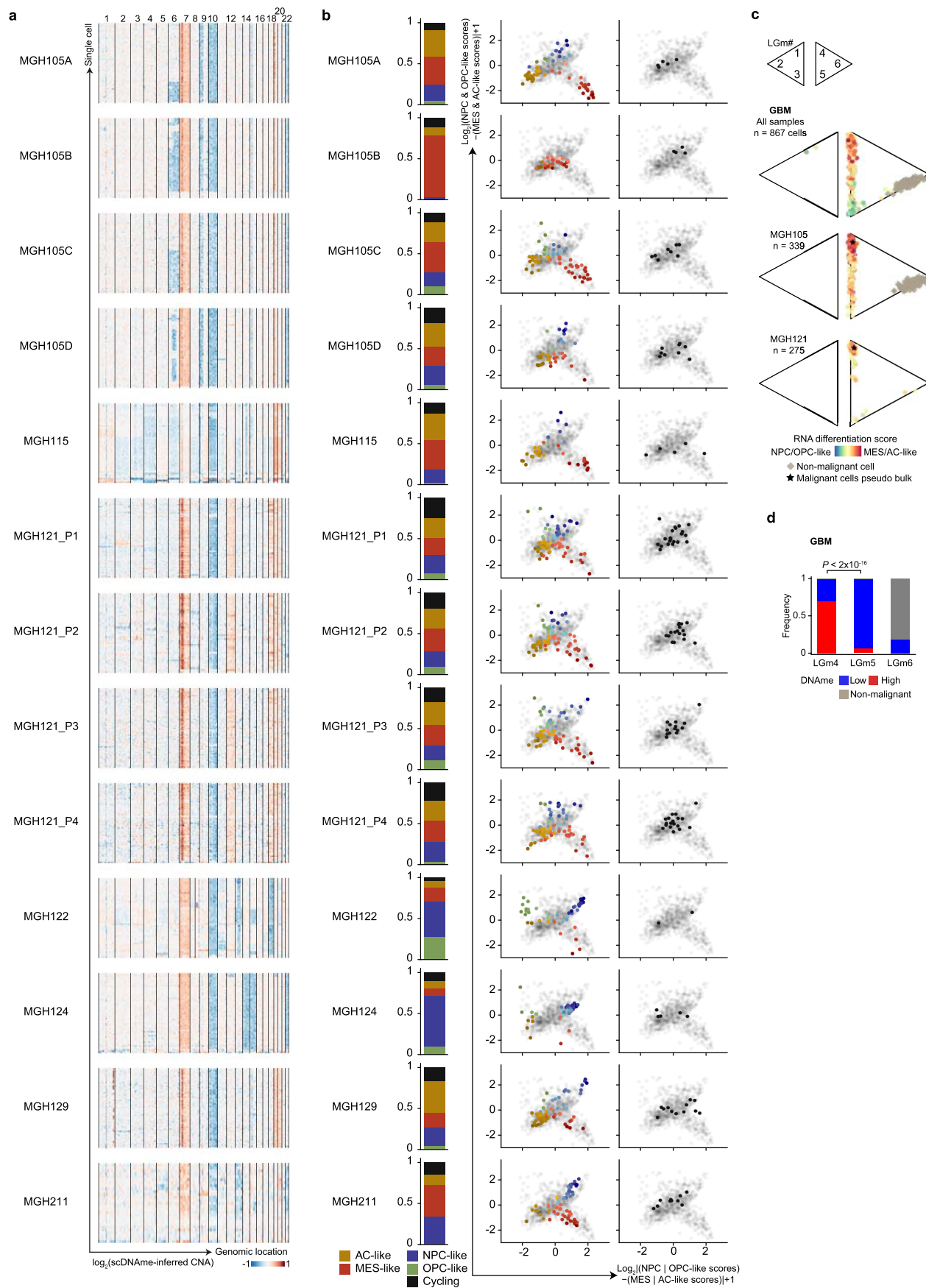
Extended data is available for this paper at <https://doi.org/10.1038/s41588-021-00927-7>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41588-021-00927-7>.

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Peer review information *Nature Genetics* thanks the anonymous reviewers for their contribution to the peer review of this work.

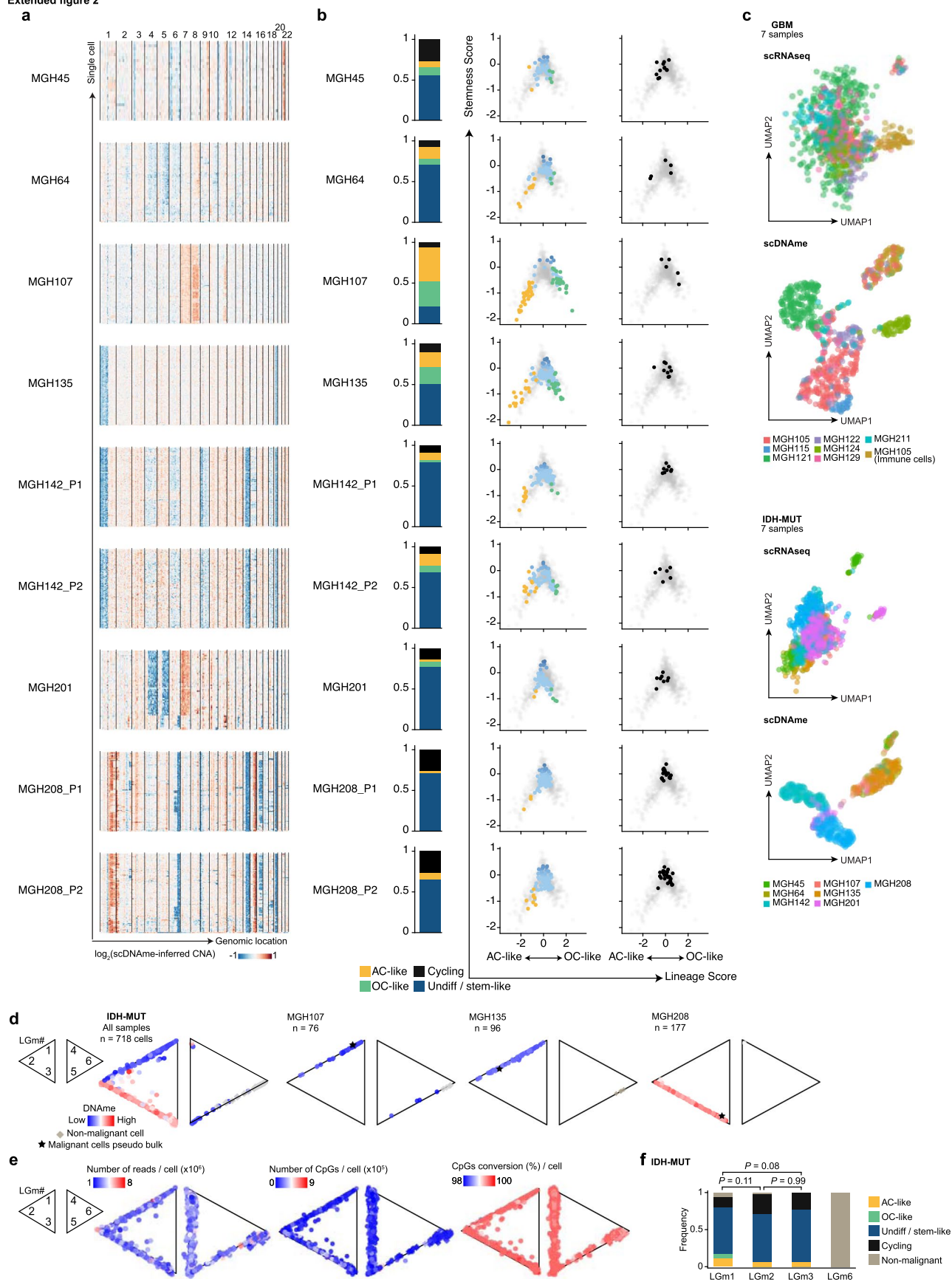
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Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Multi-omics single-cell sequencing of GBM reveals intra-tumoral DNAm heterogeneity. **a**, CNA inference based on coverage depth imbalance in the scDNAm data in windows of 20 Mb (sliding window of 5 Mb). Rows correspond to cells, clustered by overall CNA pattern. **b**, Proportion of single cells belonging to GBM cellular states (*left*) and two-dimensional representation of GBM cellular states (*middle*) or cycling cells based on the relative expression of gene-sets associated with G1.S and G2.M (*right*) for each GBM patient sample (including MGH105 biological replicates and MGH121 technical replicates). Each quadrant corresponds to one cellular state and the exact position of malignant cells (dots) reflect their relative scores for pairs of gene modules previously defined in scRNAseq data²¹. Light grey dots in the background represent all GBM samples ($n = 844$ malignant-only cells that passed quality control based on scRNAseq). **c**, Two-dimensional representation of single cells assigned to previously described LGm classes³⁹, visualized as triangle plots (where each vertex corresponds to one LGm class) across all 7 GBM samples ($n = 867$ cells [malignant and non-malignant] that passed quality control based on scDNAm, *top*), and the two samples harboring the highest number of cells: MGH105 ($n = 339$, *middle*) and MGH121 ($n = 275$, *bottom*). RNA differentiation score (defined as the difference in gene module scores between AC-/MES-like and NPC-/OPC-like cells) is overlaid. **d**, Proportion of GBM cells ($n = 867$ cells [malignant and non-malignant]) with high or low DNAm (defined as above or below the median of mean DNAm across windows of 1,000 bp around 450K array probes from TCGA glioma samples used in the analysis, respectively; Methods) assigned to previously described LGm classes³⁹. P value was determined by two-sided Fisher's exact test (**d**).

Extended figure 2

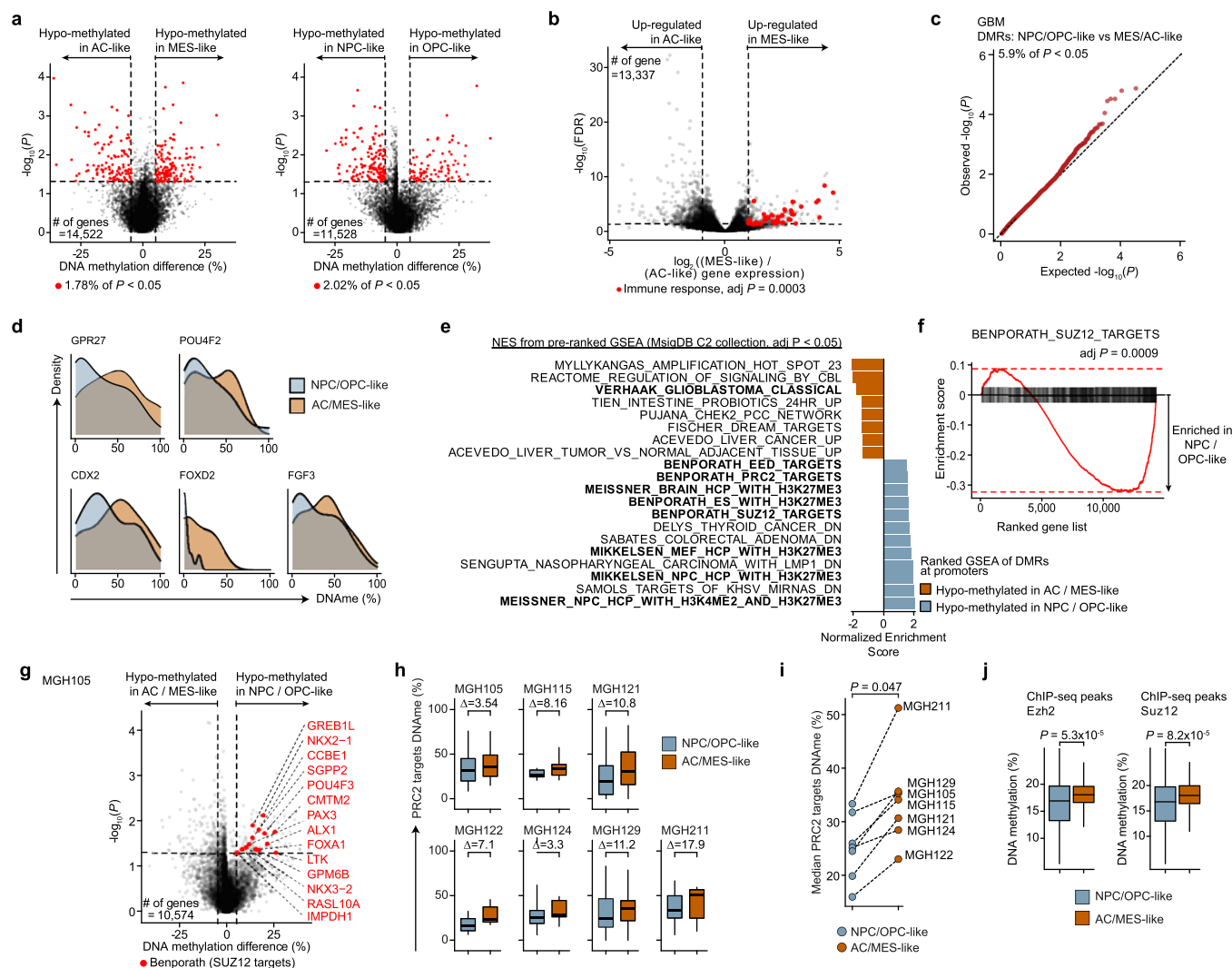


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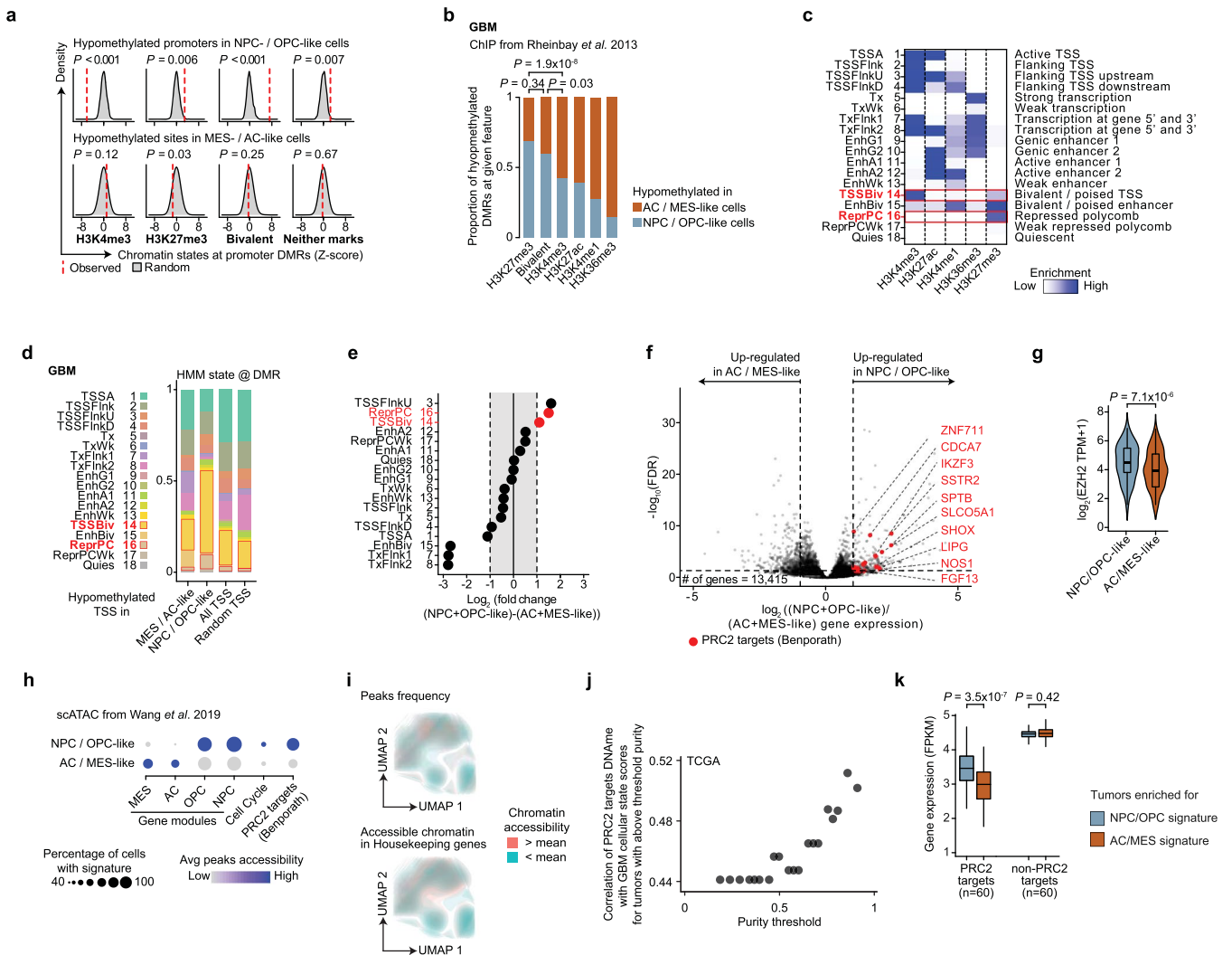
Extended Data Fig. 2 | Multi-omics single-cell sequencing of IDH-MUT reveals intra-tumoral DNAm heterogeneity. **a**, CNA inference based on coverage depth imbalance in the scDNAm data in windows of 20 Mb. Rows correspond to cells, clustered by overall CNA pattern. **b**, Proportion of single cells belonging to IDH-MUT cellular states (*left*) and developmental hierarchy representation of IDH-MUT cellular states (*middle*) or cycling cells based on the relative expression of gene-sets associated with G1.S and G2.M (*right*) for each IDH-MUT patient sample. Lineage and stemness scores define the exact position of malignant cells (dots) as computed from scRNAseq data. Light grey dots in the background represent all IDH-MUT samples ($n = 739$ malignant-only cells that passed quality control based on scRNAseq). **c**, UMAP of all single cells that passed quality control based on scRNAseq (GBM $n = 937$, IDH-MUT $n = 809$) or scDNAm (GBM $n = 867$, IDH-MUT $n = 718$). Each patient sample is indicated. See also Fig. 1b. **d**, Two-dimensional representation of single cells assigned to previously described LGm classes³⁹, visualized as triangle plots (where each vertex corresponds to one LGm class) across all 7 IDH-MUT samples ($n = 718$ cells [malignant and non-malignant] that passed quality control based on scDNAm, *left*), and three representative samples: MGH107 ($n = 76$), MGH135 ($n = 96$), and MGH208 ($n = 177$). DNAm value is overlaid. **e**, Same as (**d**) for the 7 GBM and 7 IDH-MUT samples ($n = 867$ cells [malignant and non-malignant]; IDH-MUT, $n = 718$ cells [malignant and non-malignant] that passed quality control based on scDNAm). Number of reads per cell (*left*), number of CpGs per cell (*middle*), and CpG conversion rate per cell (*right*) are overlaid. **f**, Proportion of IDH-MUT cellular states or cycling cells ($n = 718$ cells [malignant and non-malignant]) assigned to previously described LGm classes³⁹. *P* values were determined by two-sided Fisher's exact test (**f**).



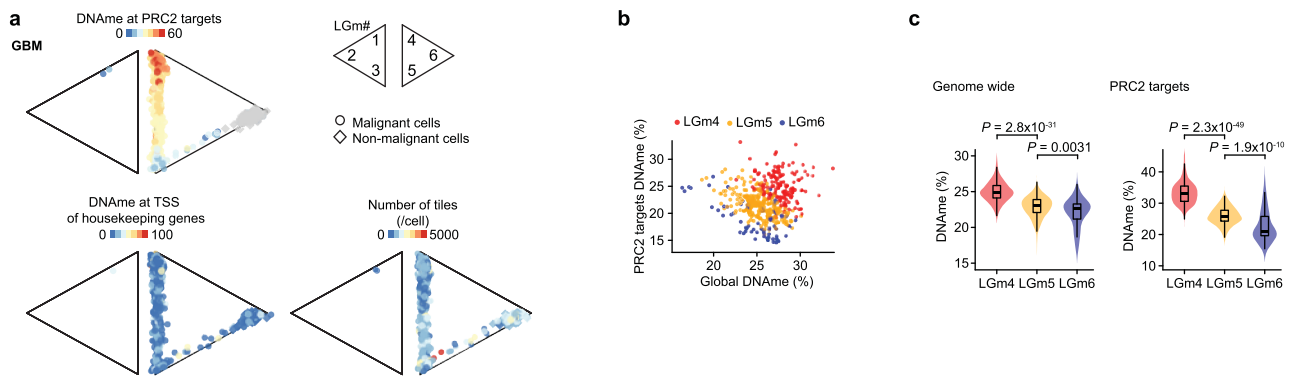
Extended Data Fig. 3 | High-resolution copy number alteration mapping enabled by single-cell multi-omics. **a**, UMAP of single cells that passed quality control based on scRNAseq (GBM $n = 937$, IDH-MUT $n = 809$). **b**, CNA inference based on bulk WES for GBM samples MGH105A/B/C, MGH122, and MGH124. *EGFR* locus is highlighted. **c**, CNA inference by scDNAme (red line) and scRNAseq (grey line) performance in correctly classifying chr. loss vs. neutral, as assessed by the AUC of ROC curve at different genomic window resolutions. ROC curve at 20 Mb resolution is shown (*inset*). 95% confidence intervals were generated using bootstrapping. **d**, CNA inferred by scDNAme (*left*) and scRNAseq (*right*) at a 50 Mb region centered at *EGFR* locus. Mean CNA profile per sample is shown in black. Red lines represent CNA segments identified by circular binary segmentation (CBS) analysis. **e**, *EGFR* expression as assessed by scRNAseq for each GBM patient sample ($n = 844$ malignant-only cells that passed quality control based on scRNAseq). **f**, Same as (**d**) for CNA inference by scDNAme at a 2 Mb region centered at *EGFR* locus. Individual cell CNA profiles are shown in grey. **g**, UMAP of single cells as defined in (**a**). Clonal chr. 7 gain (*left*) and chr. 10 loss (*middle*), as inferred by scDNAme, along with sub-clonal loss of chr. 6 (*right*), are indicated. **h**, Percentage of CpG methylation change at copy number gain, loss, and neutral chromosomal regions when comparing DNAme level of individual malignant cells to baseline for GBM ($n = 7$) and IDH-MUT ($n = 3$) samples. **i**, Same as (**h**) across all GBM and IDH-MUT samples for different thresholds adopted to define copy number gain vs. loss genomic window resolutions. *P* values were determined by two-sided Mann-Whitney U-test (**d-f**, **h-i**), comparing the *EGFR* expression median values across samples (**e**). Boxplots represent the median, bottom and upper quartiles, whiskers correspond to 1.5 times the interquartile range.



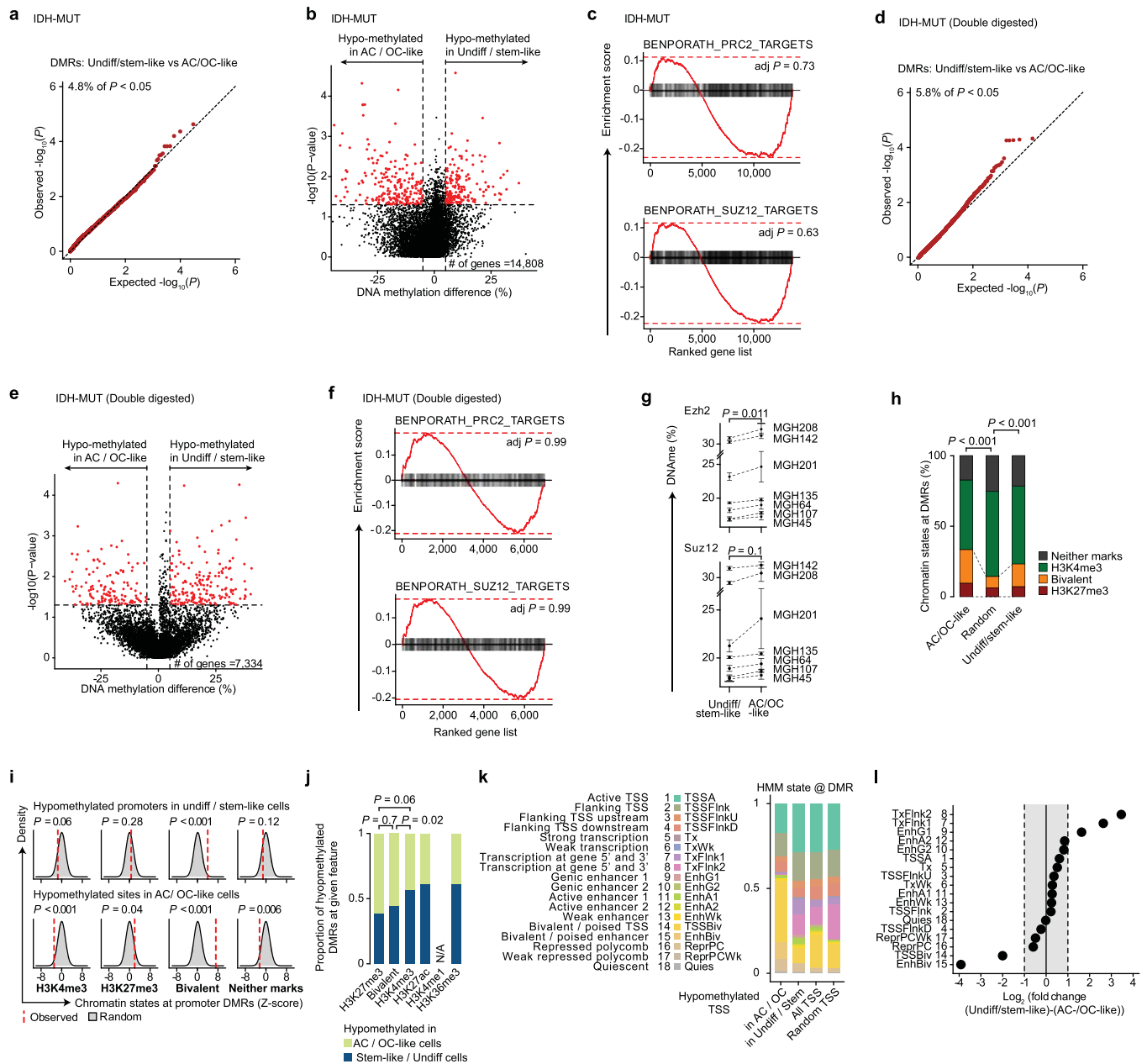
Extended Data Fig. 4 | GBM stem-like states exhibit PRC2 target hypomethylation compared with more differentiated-like cell states. **a**, Differentially methylated TSS ($\pm 1\text{Kb}$) between stem-like (NPC-like, $n = 175$ vs. OPC-like, $n = 51$; left) and differentiated GBM cellular states (MES-like, $n = 201$; AC-like, $n = 168$; right). **b**, Differential gene expression between AC-like ($n = 205$) and MES-like cells ($n = 232$). Genes with an absolute $\log_2(\text{fold-change}) > 1$ and BH-FDR < 0.05 were defined as differentially expressed (DE). DE genes belonging to immune response pathways are highlighted. **c**, Q-Q plot comparing the observed $-\log_{10}P$ values of all genes used in the differential methylation analysis between GBM cellular states (Fig. 2c) to expected $-\log_{10}P$ values. **d**, Distribution of mean promoter DNAm values in stem-like and differentiated cells for representative differentially methylated PRC2 target genes (Fig. 2c). **e**, Normalized enrichment scores for gene sets (MSigDB C2) enriched at hypomethylated promoters in NPC/OPC-like (turquoise) or MES-/AC-like (orange) cells (Fig. 2c; $n = 15,218$ genes). **f**, Enrichment score plot for SUZ12 targets⁴⁶ gene set enriched at hypomethylated promoters in NPC-/OPC-like cells (Fig. 2c; $n = 15,218$ genes). **g**, Same as (a) for a representative GBM sample (MGH105; NPC-/OPC-like, $n = 50$ cells; MES-/AC-like, $n = 138$ cells). Genes belonging to PRC2 targets⁴⁶ are labelled. **h**, Mean CpG methylation at promoters of PRC2 targets⁴⁶ between cell states for each of the 7 GBM samples. Difference in median promoter DNAm at PRC2 targets⁴⁶ between cell states is indicated. **i**, Median promoter DNAm at PRC2 targets⁴⁶ of MES-/AC-like and NPC-/OPC-like cells for each of the 7 GBM samples. **j**, Mean CpG methylation at ChIP-seq maps⁵⁰ of EZH2 and SUZ12 between GBM cell states ($n = 706$ cells). P values were determined by generalized linear model (a, c, g), weighted F-test (b), permutation test (f), two-sided Mann-Whitney U test (i, j). Boxplots represent the median, bottom and upper quartiles, whiskers correspond to 1.5 times the interquartile range.



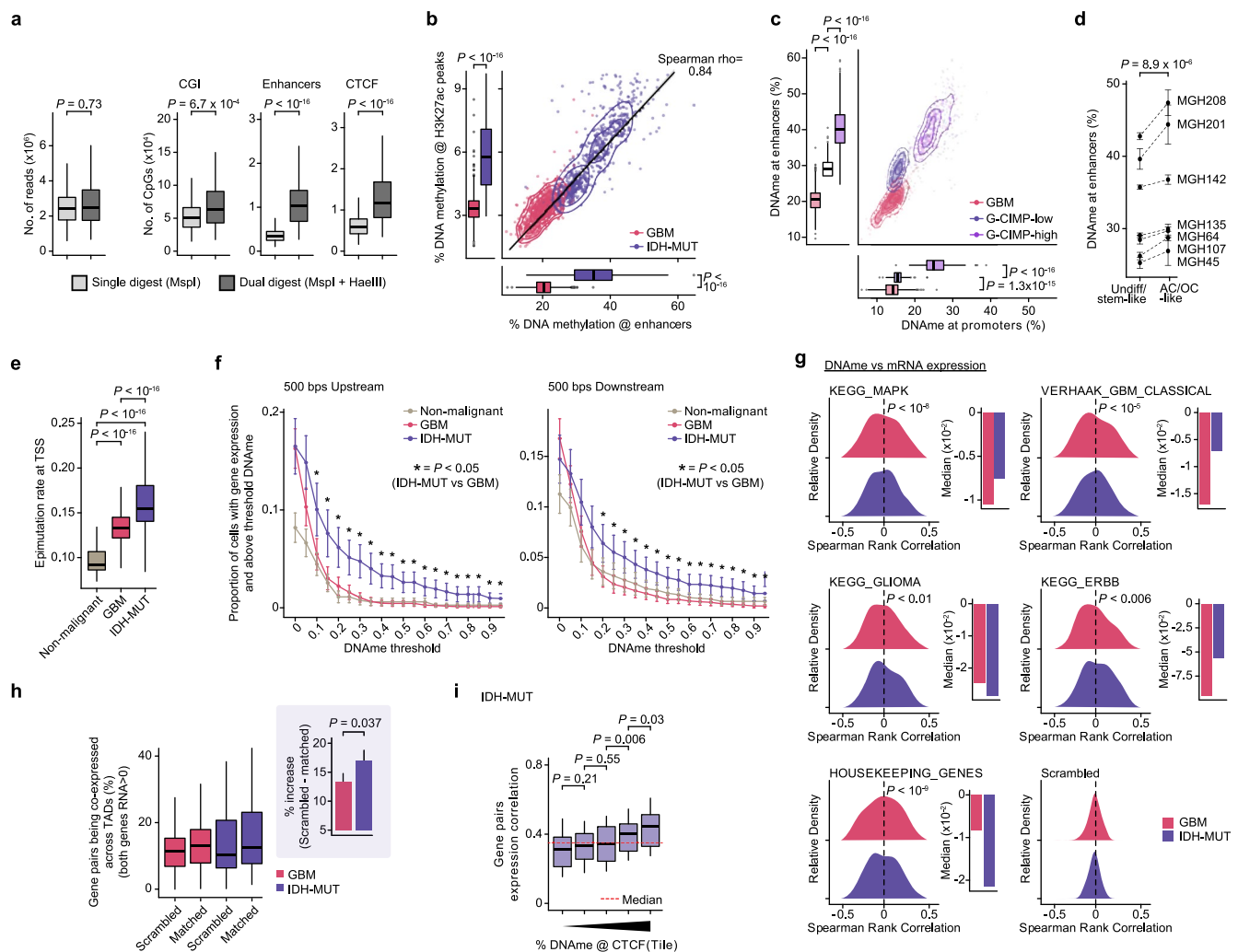
Extended Data Fig. 5 | Validation of PRC2 hypomethylation in GBM stem-like states using histone marks, single-cell ATACseq and TCGA bulk data. a, Proportion of chromatin states at randomly sampled promoters (1,000 random samplings) and hypomethylated promoters in GBM stem-like (top) vs. AC/MES-like (bottom) cells. **b**, Proportion of ChIP-seq peaks⁴⁷ at hypomethylated promoters in GBM stem-like vs. AC/MES-like cells. **c**, Heatmap of emission parameters for a HMM 18-state model derived from GBM ChIP-seq maps⁴⁷. Chromatin states of interest are highlighted in red. **d**, Proportion of chromatin states (see **c**) at hypomethylated promoters in GBM stem-like and AC/MES-like cells (Fig. 2c), all genes used in differential methylation promoter analysis ($n = 15,218$ genes), and randomly sampled promoters. **e**, Fold-change ($\log_2$) of chromatin states (see **c**) between hypomethylated promoters in GBM stem-like vs. AC/MES-like cells. Chromatin states of interest are highlighted in red. **f**, Differential gene expression between NPC-/OPC-like ($n = 270$) and AC-/MES-like cells ($n = 437$). PRC2 target⁴⁶ genes are highlighted. **g**, EZH2 expression (scRNAseq) between NPC-/OPC-like and MES-/AC-like cells across GBM samples. **h**, Gene expression activity derived from scATAC-seq open chromatin for GBM cellular states, cell cycle-related genes, and PRC2 targets⁴⁶ at distinct NPC-/OPC-like and AC-/MES-like clusters identified based on scATACseq GBM data⁵⁵. **i**, UMAP of scATACseq GBM data⁵⁵ (sample SF11956) overlaid with density plot of peaks frequency (top) and chromatin accessibility of housekeeping genes¹ (bottom). **j**, Spearman's rank-order correlation between mean DNAm at promoters of PRC2 targets⁴⁶ and RNA differentiation score and bulk sample purity for 67 TCGA GBM samples^{40,41}. **k**, Mean gene expression of hypomethylated PRC2 targets in stem-like cells ($n = 60$; Fig. 2c) and randomly selected non-PRC2 targets ($n = 60$) in TCGA GBM samples^{40,41} enriched for NPC-/OPC-like vs. AC-/MES-like signature. P values were determined by permutation test (**a**), two-sided Fisher's exact test (**b**), weighted F-test (**f**), two-sided Mann-Whitney U test (**g**, **k**). Boxplots represent the median, bottom and upper quartiles, whiskers correspond to 1.5 times the interquartile range.



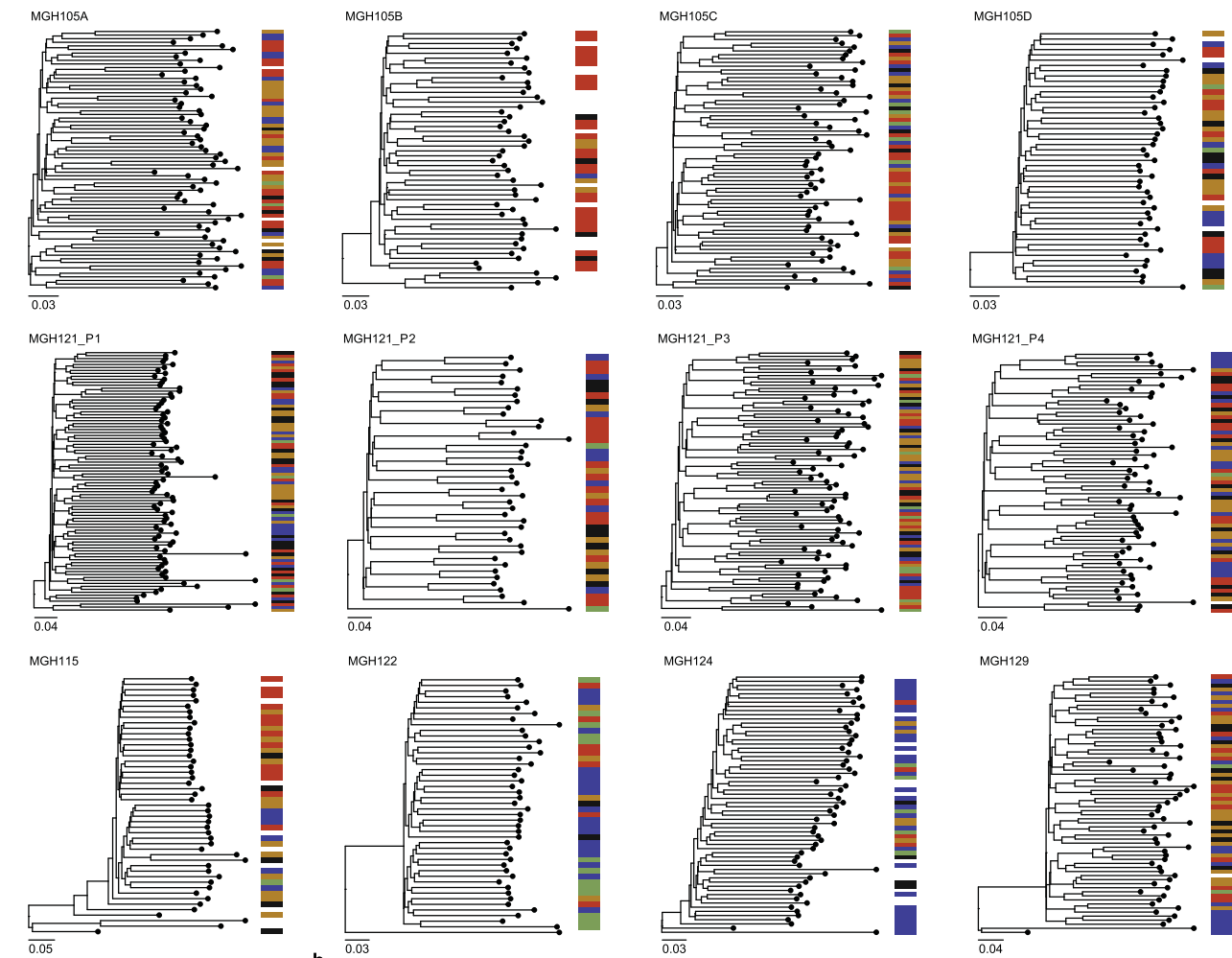
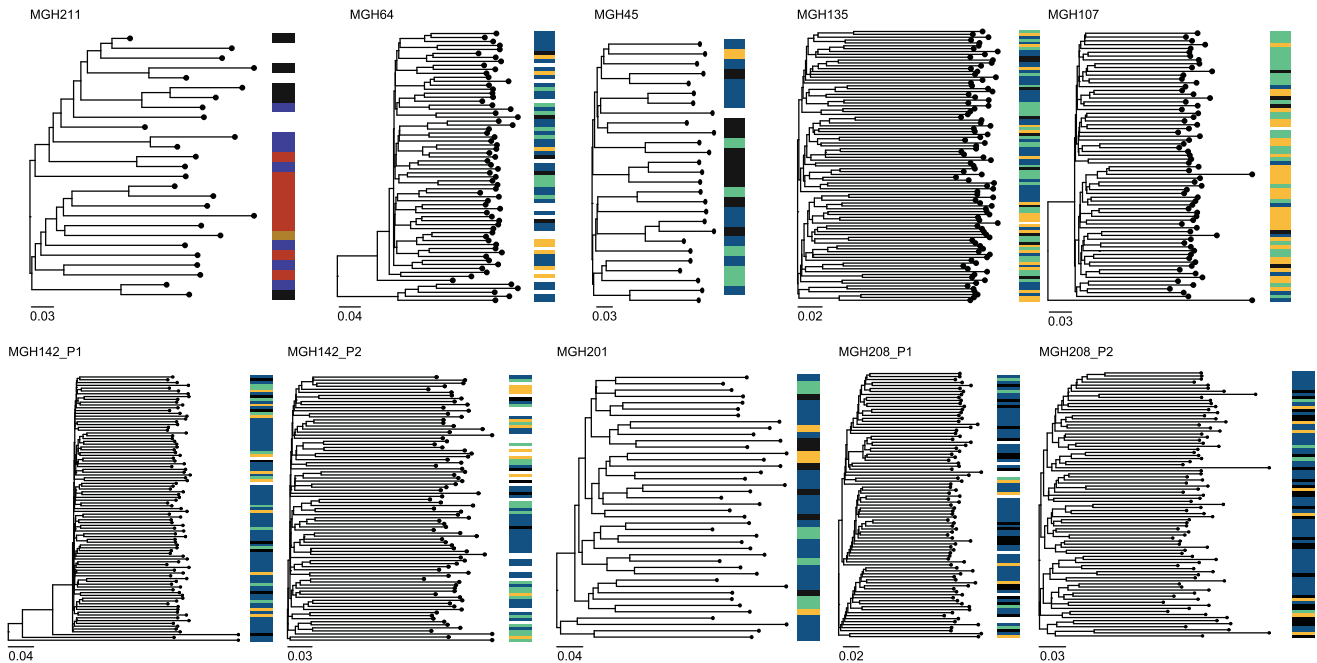
Extended Data Fig. 6 | PRC2 target DNAm underlies the classification of GBM tumors by bulk DNAm. **a**, Two-dimensional representation of single cells assigned to previously described LGm classes³⁹, visualized as triangle plots (where each vertex corresponds to one LGm class) across 7 GBM samples ($n = 867$ cells [malignant and non-malignant] that passed quality control based on scDNAm). Mean DNAm at promoters of PRC2 targets⁴⁶ (top), mean DNAm at promoters of housekeeping genes¹, and number of tiles per cell (bottom) are overlaid for each triangle plot. **b**, Comparison between mean genome wide DNAm (defined as the mean DNAm across windows of 1,000 bp around 450K array probes, Methods) and mean DNAm at promoters (TSS $\pm$ 1Kb) of PRC2 targets⁴⁶ for the 478 TCGA GBM samples that were classified as LGm4-6 by Ceccarelli *et al.*³⁹. LGm classes assignment for each sample is shown. **c**, *Left*: mean genome wide DNAm for TCGA GBM samples ($n = 478$) previously classified as either LGm4, LGm5, or LGm6 by Ceccarelli *et al.*³⁹ *Right*: mean DNAm at promoters (TSS $\pm$ 1Kb) of PRC2 targets⁴⁶ for TCGA GBM samples ($n = 478$) previously classified as either LGm4, LGm5, or LGm6 by Ceccarelli *et al.*³⁹. P values were determined by two-sided Mann-Whitney U test (**c**). Boxplots represent the median, bottom and upper quartiles, whiskers correspond to 1.5 times the interquartile range.



Extended Data Fig. 7 | Comparison of DNA methylation and chromatin state patterns between transcriptional cell states in IDH-MUT. a, Q-Q plot comparing the observed $-\log_{10}P$ values of genes used in the differential methylation analysis of promoters ($n = 14,808$ genes) between undiff/stem-like and AC-/OC-like IDH-MUT cellular states (defined in **(b)**) to expected $-\log_{10}P$ values. **b**, Differentially methylated promoters between undiff/stem-like ($n = 251$) and AC-/OC-like ($n = 133$) cells with matched scRNAseq and scDNAme data across IDH-MUT samples. Promoters with absolute mean DNAm difference $> 5\%$ and P values < 0.05 were defined as differentially methylated (red). **c**, Enrichment score plots ($n = 14,808$ genes, as in **(b)**) for PRC2 and SUZ12 targets⁴⁶ between stem-like/undifferentiated cells and AC-/OC-like cells in IDH-MUT samples. **d-f**, Same as **(a-c)**, for single-cell DNA methylomes obtained performing double digestion with HaeIII+MspI on cells from two IDH-MUT samples (MGH201 and MGH208). **g**, Mean ($\pm$ s.e.m.) CpG methylation at ChIP-seq maps⁵⁰ of EZH2 and SUZ12 between undiff/stem-like and AC-/OC-like cells in each IDH-MUT sample. **h**, Proportion of chromatin states at hypomethylated promoters in IDH-MUT AC-/OC-like cells (defined in **(b)**), randomly sampled promoters (1,000 random samplings), and hypomethylated promoters in IDH-MUT undiff/stem-like (defined in **(b)**). **i**, Proportion of chromatin states at randomly sampled promoters (1,000 random samplings) and hypomethylated promoters in IDH-MUT undiff/stem-like (top) vs. AC-/OC-like cells (bottom). **j**, Proportion of ChIP-seq peaks⁴⁷ at hypomethylated promoters in IDH-MUT undiff/stem-like vs. AC-/OC-like cells (defined in Extended Data Fig. 5c) at hypomethylated promoters in IDH-MUT undiff/stem-like (defined in **(b)**), hypomethylated promoters in IDH-MUT AC-/OC-like cells (defined in **(b)**), all genes used in differential methylation promoter analysis ($n = 14,808$ genes), and randomly sampled promoters, respectively. **l**, Fold-change ($\log_2$) of chromatin states between hypomethylated promoters in IDH-MUT undiff/stem-like vs. AC-/OC-like cells. P values were determined by generalized linear model (**a-b, d-e**), Fisher's combined probability test (**g**), permutation test (**c, f, h-i**), two-sided Fisher's exact test (**j**).



Extended Data Fig. 8 | IDH-MUT cells exhibit preferential enhancer hypermethylation, decoupling of the promoter methylation-expression relationship and disruption of CTCF insulation. **a**, Number of aligned reads and unique CpGs for MspI ($n=476$) and HaeIII+MspI digested IDH-MUT cells ($n=242$; MGH201 and MGH208). **b**, Mean CpG methylation at FANTOM5 enhancers vs. H3K27ac ChIP-seq peaks^{47,70} between GBM ($n=765$) and IDH-MUT ($n=670$) cells. **c**, Mean CpG methylation at TSS (± 1 kb) vs. FANTOM5 enhancers between GBM ($n=765$) and IDH-MUT ($n=670$) cells (G-CIMP-low [MGH107, MGH135, MGH45, MGH64]; G-CIMP-high [MGH142, MGH201, MGH208]). **d**, Mean ($\pm$ SEM) CpG methylation at FANTOM5 enhancers for stem-like/undifferentiated and AC-/OC-like IDH-MUT cells. **e**, Epimutation rate across non-malignant ($n=148$), GBM ($n=765$) and IDH-MUT ($n=670$) cells. **f**, Proportion of cells with gene expression (read count >0) and above-threshold DNAm at 500 base-pairs regions upstream (left) or downstream (right) of TSS. Data are mean ($\pm$ s.e.m.) across all genes (expression seen in >5 cells, DNAm >5 CpGs per region) for non-malignant cells ($n=148$), GBM ($n=765$) and IDH-MUT ($n=670$) cells. $*P$ -value < 0.05 . **g**, Left: Distribution of Spearman's rho of expression and promoter DNAm correlation ($n=1,523$ genes expressed >5 cells, DNAm >5 CpGs per promoter); GBM ($n=765$) and IDH-MUT ($n=670$) cells. Right: Median values of Spearman's rho of expression and promoter DNAm correlation. **h**, Percentage of genes pairs across CTCF sites⁷⁰ being co-expressed (both RNA read count >0); GBM ($n=765$) and IDH-MUT ($n=670$) cells. Scrambled represents randomly permuted cell labels for the expression values. Inset: Increase in percentage of genes pairs across CTCF sites⁷⁰ being co-expressed when comparing matched vs. scrambled groups. Error bars represent 95% CIs. **i**, Gene expression correlation (Spearman's rho) of genes pairs across CTCF sites⁷⁰ per tile of mean CpG methylation at CTCF binding sites (low-to-high); IDH-MUT ($n=670$) cells. P values are two-sided Mann-Whitney U test (**a-c**, **e-f**, **h-i**), Fisher's combined probability test (**d**), two-sided Kolmogorov-Smirnov test (**g**). Boxplots represent the median, bottom and upper quartiles, whiskers correspond to 1.5 times the interquartile range.

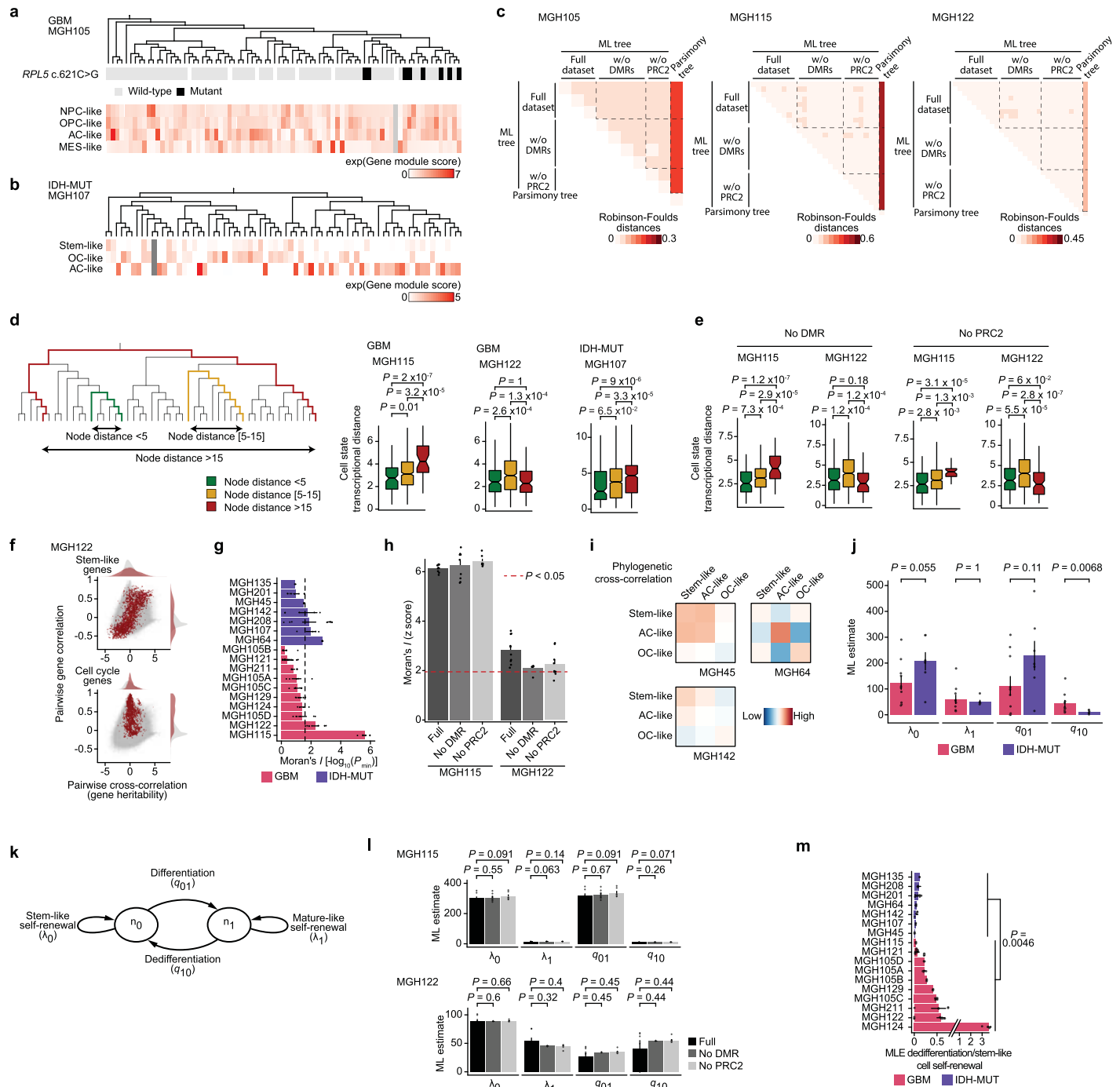
a GBM samples:**b** IDH-MUT samples:

GBM samples: NPC-like AC-like Cycling
 OPC-like MES-like NA

IDH-MUT samples: AC-like Undiff / stem-like
 OC-like Cycling NA

Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | High-resolution DNAm-based lineage trees coupled with leaf annotation of cellular states. a, Representative (random cell subsampling) DNAm-based lineage tree for each GBM patient sample (including MGH105 biological replicates and MGH121 technical replicates), with projection of GBM cellular states. **b,** Representative (random cell subsampling) DNAm-based lineage tree for each IDH-MUT patient sample (including MGH142 and MGH208 technical replicates), with projection of IDH-MUT cellular states. Throughout the figure, scale represents DNAm changes per site.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Cell state transition dynamics inference from lineage tree architectures revealed higher cellular plasticity in GBM compared to a more stable differentiation hierarchy in IDH-MUT. **a**, *Top*: GBM DNAME-based lineage tree (MGH105) with *RPL5* c.621 C>G genotyping. *Bottom*: GBM gene module scores. **b**, IDH-MUT DNAME-based lineage tree (MGH107) with IDH-MUT gene module scores. **c**, Normalized Robinson-Foulds between GBM tree replicates (from same sample; full dataset or removing CpGs from DMRs (Fig. 2c) or PRC2 targets⁴⁶) reconstructed by maximum-likelihood (ML) vs. maximum parsimony. **d**, Transcriptional distances as function of lineage distance between unique cell pairs for MGH115, MGH122 and MGH107. **e**, As (**d**), for DNAME-based lineage tree of MGH115 and MGH122 ($n=47$ and 46 cells, respectively) reconstructed removing CpGs from DMRs (Fig. 2c) or PRC2 targets⁴⁶. **f**, Pairwise gene expression correlation (Pearson's) and cross-correlation (heritability). Grey points=all gene pair relationships; red points=gene pair relationships within selected gene module (*top*: stem-like; *bottom*: cell cycle). **g**, Phylogenetic association of cell states on GBM ($n=7$ patients; $n=10$ samples with MGH105A-D) and IDH-MUT ($n=7$ patients). Barplots=weighted mean $\pm$ s.e.m. Moran's I permutation-based one-sided P values (10^6 permutations) across replicates. Dashed line: $P=0.025$. **h**, As (**g**), comparing DNAME-based lineage tree reconstruction of MGH115 and MGH122, using replicates from same sample with full dataset or removing CpGs from DMRs (Fig. 2c) or PRC2 targets⁴⁶. Barplots=mean $\pm$ s.e.m. **i**, Heat maps of pairwise cell state phylogenetic associations. Close phylogenetic associations are shown in warmer colors. **j**, ML estimate (median $\pm$ MAD across tree replicates; samples as in (**g**)) rates of cell state growth and transition. **k**, Mathematical model of glioma evolutionary dynamics. **l**, ML estimate (mean $\pm$ s.e.m. across tree replicates of MGH115 and MGH122) rates of cell state self-renewal and transition, using replicates from same sample (full dataset or removing CpGs from DMRs analysis (Fig. 2c) or PRC2 targets⁴⁶). **m**, Weighted median $\pm$ weighted MAD rates of dedifferentiation compared to stem-like cell self-renewal across lineage tree replicates (sample as in (**g**)). P values: two-sided Mann-Whitney U test (**d-e**, **j**, **l-m**). Boxplots: median, bottom and upper quartiles, whiskers: 1.5 times the interquartile range.

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| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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Software and code

Policy information about [availability of computer code](#)

Data collection No software was used for data collection.

Data analysis Bismark (v.0.14.5); STAR(version 2.5.2a); ZINB-WaVE(v. 1.6); edgeR (v. 3.26.8); BEDTools (v2.27.1); T-Gene (MEME v5.1.1); fgsea (v.10.1); scanpy (v1.4.4); RSEM (v1.3.1); DAVID (v. 6.8); DNACopy (v1.60.0); InferCNV (v1.2.1) ; scikit-learn (v0.21.3); Cell Ranger (v.3.1); Signac (v1.1.0); LOLA (v1.16.0); BWA v0.7.17; MACS2 (v2.0.10); ChromHMM (v1.20); IQ-TREE v1.6.9; FigTree v1.4.3; diversitree (v0.9.15); phytools (v0.7.70); GenSA (v1.1.7); bbmle (v1.0.23.1); Seurat (v3.9.9.9038); SeuratWrappers (<https://github.com/satijalab/seurat-wrappers>); velocyto.R (v0.6); Python 3.0; R version 3.6.1; FACSDiva software v8.0.1

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Processed data generated for this study are available through the NCBI Gene Expression Omnibus (GEO) under accession number GSE151506. Raw data access can be requested through the Data Use Oversight System (DUOS) Dataset Catalog with dataset ID of DUOS-000133 as well as through the European Genome-phenome Archive (EGA) with dataset ID of EGAS00001005472. The data can be visualized and interrogated through the Broad Institute Single-Cell Portal: https://singlecell.broadinstitute.org/single_cell/study/SCP936. Single-cell ATAC-seq data is available under the EGA repository EGAS00001002185, EGAS00001001900, and

EGAS00001003845 and NCBI GEO accession number GSE138794. The Cancer Genome Atlas (TCGA) data (DNA methylation, gene expression and clinical profiles) is available under TCGA database (<https://cancergenome.nih.gov/>). ChIP-seq data is available under the NCBI GEO accession number GSE46016.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The sample size of our single-cell multi-omics profiling study was dependent on the availability of human patient tissues. We sequenced 2,304 single cells in total from 7 and 7 GBM and IDH-MUT patients, respectively, giving us enough statistical power to detect differences at the single cell level in all the analyses reported in the study (>80% power [$\beta = 0.2$] to detect differences at a 5% significance level [$\alpha = 0.05$] using a two-sided test).
Data exclusions	No data were excluded from the study.
Replication	We performed 3+ biological replicates for each glioma subtype, by generating multi-omics single-cell profiling, integrating DNA methylation (DNAm), transcriptome and genotyping of 2,304 cells from 7 GBM and 7 IDH-MUT patient samples. 4 technical replicates of GBM patient sample MGH121 were generated; 2 technical replicates of IDH-MUT patient MGH142, MGH208 and MGH201 were generated. All attempts at replication were successful.
Randomization	Randomization is not applicable as no experimental groups were used in our study.
Blinding	Blinding is not applicable as no experimental groups were used in our study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Anti-human CD45-Vioblue direct antibody conjugate (clone REA747, Miltenyi Biotec; Cat#130-110-775; RRID: AB_2658242) and anti-human CD3-PE direct antibody conjugate (clone BW264/56, Miltenyi Biotec; Cat#130-113-691; RRID:AB_2726232). Antibodies were diluted according to the manufacturer's recommendations.
Validation	All antibodies used were validated for their use in FACS experiments with human samples, as shown on the website provided by the company.

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics	This study analyzes primary human glioma cells resected from patients. The covariate-relevant population characteristics of the human research participants are provided as Supplementary Table 2.
Recruitment	Adult patients operated on for a glioma at Massachusetts General Hospital (MGH), or Dana-Farber Cancer Institute (DFCI) who consent to have tissue used for research purposes. The collection of tissue samples at MGH and DFCI was approved

through DFCI IRB protocol #10-417 through Office for Human Research Studies (OHRS). There were no specific recruitment methodologies. No self-selection biases were present during recruitment. Patients who were operated on for a glioma at the MGH or DFCI (and when appropriate their legal representative) were approached by the neurosurgeon for participation.

Ethics oversight

All adult patients included in this work provided preoperative informed consent to take part in the study according to the Institutional Review Board protocol Dana-Farber/Harvard Cancer Center 10-417.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Fresh tumor specimens were collected on phosphate buffered saline (Gibco, Waltham, MA) and mechanically dissociated into small, 0.5-1mm, pieces with a disposable, sterile scalpel and further dissociated into single cell suspensions using the enzymatic brain tumor dissociation kit (P) from Miltenyi Biotec (Bergisch Gladbach, Germany), following the manufacturer's protocol. Cell suspensions were subsequently stained for flow cytometry using antibodies specific for anti-human CD45-Vioblu direct antibody conjugate (clone REA747, Miltenyi Biotec; Cat#130-110-775; RRID: AB_2658242) and anti-human CD3-PE direct antibody conjugate (clone BW264/56, Miltenyi Biotec; Cat#130-113-691; RRID:AB_2726232). Antibodies were diluted according to the manufacturer's recommendations. Tumor cell suspensions were next spiked with 0.5 μ M Calcein AM (ThermoFisher, Waltham, MA) and 1 μ M TO-PRO-3 (ThermoFisher) to enable gating of live cells (Calcein AM+, TO-PRO-3-) and incubated in the dark at room temperature for 10 minutes. Live, single cells from all GBM and IDH-mutant gliomas were identified by size (forward scatter laser light), granularity (90 degree side scatter laser light) and specific stains (gating: Calcein AM+, TO-PRO-3-, CD45-; see Supplementary Figure 1). Viable single cells were sorted into individual wells of a 96-well twin.tec PCR plate (Eppendorf; Hamburg, Germany) that contained 10 μ L/well of TCL buffer (Qiagen; Venlo, Netherlands) with 1% beta-mercaptoethanol, using a BD Biosciences (Franklin Lakes, NJ) Fusion fluorescence-activated cell sorter (FACS) with a 70 μ M nozzle and running FACSDiva software v8.0.1.

Instrument

BD Biosciences (Franklin Lakes, NJ) Fusion fluorescence-activated cell sorter (FACS) with a 70 μ M nozzle

Software

FACSDiva software v8.0.1

Cell population abundance

After dissociation, glioma samples were composed of tumor cells and immune cells from the tumor micro-environment (TME). In general, we observed 30-40% tumor cells with the complement being TME cells.

Gating strategy

Live, single cells from all GBM and IDH-mutant gliomas were identified by size (forward scatter laser light), granularity (90 degree side scatter laser light) and specific stains (gating: Calcein AM+, TO-PRO-3-, CD45-).

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.